

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/371911867>

Connections between Dimensional Analysis, Mathematical Logic and Number Theory: How powerful can Dimensional Analysis be?

Preprint · June 2023

DOI: 10.13140/RG.2.2.29973.91367

CITATIONS

0

READS

68

1 author:



Juan Antonio Martínez Rojas

University of Alcalá

36 PUBLICATIONS 399 CITATIONS

SEE PROFILE

Connections between Dimensional Analysis, Mathematical Logic and Number Theory: How powerful can Dimensional Analysis be?

Juan Antonio Martínez Rojas*

*Department of Signal Theory and Communications,
Universidad de Alcalá, Escuela Politécnica,
Campus Universitario, Alcalá de Henares (Madrid), Spain*

Abstract

Dimensional Analysis is related with many mathematical branches. Perhaps one of the most unsuspected similarities is between Dimensional Analysis and Number Theory. Dimensions share many properties with prime numbers. Guided by this analogy, I have applied a Gödel numbering procedure to physical dimensions. The results of the application of number theoretic concepts to Dimensional Analysis are surprisingly rich and powerful. Several models of Physics are possible which are compatible with our current physical theories in the sense that equations are valid in all of them. However, there are subtle differences between these models and the conceptual implications are very important. One of the main results of this study is that the mapping between dimensions and primes is not completely arbitrary. In fact, if consistent interpretations of physical theories are expected, the number of arithmetic dimensional models is very restricted, but the numbering scheme is so general that it could include new undiscovered dimensions if necessary. There are two classes of dimensional frameworks for the formulation of physical theories. The first one is our current linear treatment of dimensions where only nondimensional quantities can be equaled using Rayleigh polynomials. However, the generalization of Dimensional Analysis to more general arithmetics allows the definition of modular models in which all dimensions are entangled in far more complex ways than in our present linear understanding of Physics. Measurements are interpreted as dimensional entanglements and the use of modular models allows new powerful connections among dimensions, variables and equations. This extension of Physics permits us to explore universes in which cyclic connections between physical quantities exist. The result is the conceptual unification of Physics and Mathematics in new ways. For example, striking connections among mass, charge and spin are discovered, while entropy, time and area are directly related and even some transcendental numbers can be identified with physical dimensions. Perhaps the most beautiful discovered relation is the reproduction of the Euler's formula of Complex Analysis in some dimensional modular models using a completely discrete framework. This result permits us to extend dimensions to the complex domain in full parallelism with Analytic Number Theory. Even more surprising is that this relation is fulfilled only in a few bases, which are precisely those models able to produce meaningful physical equations from complex dimensions directly. The resulting physical equations are extraordinarily compact and have a very similar structure to their geometric algebra and differential form formulations, which points to a direct geometrical interpretation of complex dimensions. If the unification of Dimensional Analysis and Number Theory is only a mere

analogy or has more profound implications is a very important question, because many results in noncyclic (linear) Physics can be reproduced by cyclic (modular) Physics, but the converse is not true. If arbitrariness in the Gödel numbering is assumed, every model would be a valid view in a multidimensional framework and several conceptual interpretations of Physics are possible and compatible. If this is true, different interpretations of Quantum Mechanics, for example, would be compatible and no one would be preferred. However, complex dimensions and their geometrical meaning only hold in three different bases for a reasonably low number of basis dimensions and only two of them are compatible with the number of spatial dimensions of our present Universe. In any case, the existence of modular models of Physics opens new interesting avenues to the unification of Physics and Mathematics and provides beautiful results that deserve further study by themselves. How a pure natural number ordering is able to produce, for example, the Maxwell equations in their most geometrical form, is as a beautiful and amazing result as it can be.

*Electronic address: juanan.martinez@uah.es

I. INTRODUCTION

In my previous work [1] I explored a new answer to the Wigner's question: *The Unreasonable Effectiveness of Mathematics in the Natural Sciences* [2] extending the framework of Dimensional Analysis (DA) to the infinitesimal domain. Now I propose to go a step further in the connection between Mathematics and Physics, exploring a direct link between Mathematical Logic, Number Theory and Dimensional Analysis. In principle, dimensions has no natural ordering. We cannot speak meaningfully about time being greater than length, or mass less than charge. Besides, basis dimensions are arbitrary. We can select any consistent set of physical variables as basis dimensions. But, if we forget all of this and impose an ordering to the basis dimensions, are there any consequences? Is it even possible to make sense of such ordering?

Inspired by fundamental results in Mathematical Logic and the Foundations of Mathematics, I have assigned a very simple Gödel numbering to physical dimensions [3]. The numbering scheme consists only in equaling each basis dimension with a prime number. In this way: length = 2, time = 3, mass = 5, temperature = 7 and charge = 11, for example. Although this seems completely arbitrary, it is not the case. This will be one of the first great surprises of the connection between Number Theory and DA.

Gödel numbering is a common technique in many results of Mathematical Logic [4], including the famous Incompleteness Theorem, Constructive Mathematics, where Markov's RUSS interpretation is the most direct application [5], and Reverse Mathematics, which uses the coding of mathematical objects as natural numbers to study the minimal sets of axioms required to prove a theorem in the context of Second Order Arithmetic [6].

Number Theory is one of the oldest mathematical disciplines and its results pervade all of Mathematics. In fact, some of the most challenging and unsolved problems in Mathematics are related with conjectures about prime numbers, like the Goldbach conjecture or the Riemann Hypothesis [7, 8]. Of course, such problems are outside the scope of this work, but I want to remark one very important fact, the deep connection between prime numbers and Complex Analysis, which has produced the powerful results of Analytic Number Theory [9]. We will see that such connection appears also in the number theoretic description of DA with crucial consequences.

This work will develop the following themes:

1. The exploration of the mapping between prime numbers and physical basis dimensions. Prime numbers and physical dimensions share many properties, to the point that a meaningful mapping between them is possible.
2. Physical variables can be expressed as operations using prime numbers. Any physical variable can be converted into a natural number using only elementary arithmetic.
3. Modular Dimensional Analysis (or Arithmetic), MDA. We can develop a completely new approach to DA using modular arithmetic. This produces new physical theories and interpretations that are compatible with our current knowledge but reveals unsuspected complex entanglements between dimensions. Our present Physics can be considered linear from the point of view of DA, but other dimensional frameworks are possible, using cyclic mathematical structures. Other interpretation is that Linear Physics is a formal infinite series formed by the sum of dimensional cyclic bases, pointing to a strong analogy with Harmonic Analysis.
4. Linear Physics uses Linear Algebra or Rayleigh monomials to verify or produce physical equations, which generates nondimensional quantities from the variables describing certain phenomena. However, MDA is able to reproduce many equations directly without the use of nondimensionalization. Even more striking is the fact that the obtained equations appear in their most geometrical form, resembling the structure they have using Clifford Algebras (Geometric Algebra) or differential forms. However, this is possible only in a very selected set of modular bases, revealing that the apparent arbitrariness of DA in Linear Physics is really rather limited. The consequences of this fact are far reaching. The results imply that MDA heavily restricts the dimensional characteristics of physical theories, if we want to reproduce the experimentally proven physical equations. Not only classical equations, like the Maxwell's or the Hamilton - Jacobi's ones are obtained, but quantum results are also revealed, even reproducing the structure of the creation and annihilation operators. I will stop at that point, because GR and QFT require a much more detailed and careful study.
5. Another interesting fact is that MDA is able to reproduce continuum results using only rational numbers. Infinitesimal and real numbers are not needed to produce meaningful physical equations. Not only this, complex numbers appear naturally and

even the Euler's formula holds in the selected bases. The unexpected consequence is that some transcendental numbers, very important in Physics and Mathematics, have dimensions in MDA. In other words, from a dimensional point of view, Physics is geometrical and Geometry is physical in a fundamental sense.

6. Noncommutativity is essential in QM and QFT, but this is impossible using only rational numbers without some kind of higher structures like matrices. However, the existence of complex modular dimensions permits the definition of noncommutative structures using only elementary arithmetic without matrices. The result is that we can reproduce quantum results using sums instead of products of variables, pointing to a simple dimensional representation of complex objects like spinors. For example, in Linear Physics the sum of different dimensions is completely forbidden and only products are meaningful. In fact, this is the very foundation of DA, all equations must be dimensionless. However, in MDA we can sum different dimensions in a rigorous way, because all of them are related and converge to the unity (nondimensionality) in a cyclic form. When the discrete description of the continuum, its complex analysis extension and the sum of dimensions are combined, QM "miraculously" emerges. The conversion of multiplications in complex sums should not sound strange, in fact, the Schönhage–Strassen algorithm [10] uses the discrete Fourier transform to perform the multiplication of large numbers very efficiently. The really remarkable result is that we can operate physical dimensions as freely as pure numbers in MDA, obtaining correct physical equations in the end.

In Mathematics, Analytical Number Theory proves that the complex continuum is deeply connected with the discrete of natural numbers. In Physics the same relation holds, the continuum Geometry use of Linear Physics corresponds to analog computation, while MDA produces the digital equivalent computation of the same functions. Even (pseudo)randomness can be simulated using modular arithmetic as Cryptography shows [11]. Also, Rayleigh monomials can be seen as braids in Linear Physics, but in Modular Physics we add rotations creating twisted braids. This is just the description of a tangloid [12], a device invented by Piet Hein to visualize spinors. Thus, in a sense, Linear Physics is a kind of dimensional tensorial representation, while Modular Physics is a spinorial description.

This investigation is a way to explore if there is a natural order in physical dimensions

or they are truly arbitrary. The coding of dimensions as prime numbers produces far richer results than could be expected, providing new tools for a research program in Reverse Physics, in the same spirit than Reverse Mathematics [6]. The main goal is to reconstruct Physics from pure Mathematics + Physical Semantics (dimensional interpretation of pure numbers). Of course, this can be seen symmetrically and say that we can reconstruct Mathematics from Physics without dimensions. Through DA, Reverse Physics can provide deep results for other branches of Physics.

Reverse Physics shares many objectives with the exploration of alternative interpretations of physical theories and the connections between classical and quantum theories (how much of quantum theory can be reproduced by classical means?). But this program is approached from a much more elemental dimensional level. A very puzzling fact is that coding of mathematical objects as numbers is a very difficult task in general in Constructive and Reverse Mathematics, but the Gödel's numbering of physical dimensions in a certain order produces very relevant results without any further coding of the underlying mathematical structures. For example, in one basis the entropy is dimensionally the same as time, but in the other privileged basis it is the modular equivalent to the imaginary unit in Complex Analysis, so that there is a direct connection between time, entropy and Quantum Mechanics. Other relevant results are, for example, that such a complex object as Maxwell's equations can be directly obtained from pure numbers without any additional hypotheses, that Euler's formula holds only for some rational approximations of its corresponding transcendental numbers in a few privileged dimensional bases, that some dimensions coincide with such transcendental numbers and that infinitesimals can be represented by rational numbers with prime denominators in the selected bases.

II. NUMBER THEORETIC DESCRIPTION OF PHYSICAL DIMENSIONS AND FOUNDATIONAL ISSUES

As previously commented, the Gödel's numbering scheme of physical dimensions is an almost trivial task. The key is the ordering, which is not really arbitrary. We can apply some rules of thumb, which sound logical, but full justification is only guaranteed by the results. The rules are:

1. Our first selection of basis dimensions is: length, time, mass, temperature and electric

charge.

2. Length is the dimension with the smallest prime number, because the number of spatial dimensions in our universe is the largest, three spatial dimensions versus one of the rest. This allows the description of area and volume inside one modular cycle.
3. The natural order of this basis is: Length, time, mass, temperature and charge.

In order to simplify the notation, I will use standard uppercase letters for dimensions instead of sans-serif ones. If other variables are used, they will be properly named to avoid confusion. A very important thing to remember is that every result will be in the context of Dimensional Analysis. Using these previous simple rules, the natural coding is:

- Length: $L = 2$.
- Time: $T = 3$.
- Mass: $M = 5$.
- Temperature: $\Theta = 7$.
- Charge: $Q = 11$.

Very few exceptions to this rule will be noted when I use a module less than 13. Using this numbering, I will write every basis in the notation: $(L2, T3, M5, \Theta7, Q11) \pmod{N}$. Our current Physics is based on a linear notion of physical dimensions. All basis dimensions are different, but they can coincide in some systems of units, like the commonly used geometrized unit system in General Relativity, where $c = 1, G = 1$, or natural units, which considers also $\hbar = 1$. However, in these systems the aim is to simplify the writing of the equations or to define some relevant scales for physical phenomena, but not a real dimensional unification. Thus, I propose to study the consequences of the unification of physical dimensions, besides the ultimate goal of unification of all physical theories. In the end, both approaches must be connected, but this connection is not obvious, so that it is better to develop this new approach to Dimensional Analysis without any further claims.

One of the most relevant differences between Physics and Mathematics is precisely that all quantities in Physics have a dimension, while mathematical structures are, in general,

dimensionless. However, if we can reproduce many, perhaps all, results of Physics using a pure number coding, we have to reconsider some important questions about the Philosophy of Physics:

- What are physical dimensions really? Are they truly arbitrary or are there some natural dimensions like length, time or mass?
- Are dimensions products of our cognitive structures in a more or less Kantian sense or are they related with real structures of our Universe?
- Is Physics really dimensionless or, on the contrary, is Mathematics fully dimensional in disguise?
- Is the existence of physical dimensions epistemological, ontological or merely semantical, as an alternative interpretation of Mathematics?
- In the end, are Physics and Mathematics truly different? In other words, would they be ontologically different even if a formal equivalence between the two could be proved?

For example, if Physics is really dimensionless, as all its equations must obey, then, in the end, physical dimensions would be also pure numbers, because we can always transform them in dimensionless numbers by changing the basis. But, if Physics is dimensionless, the only difference between Physics and Mathematics is that our Universe restricts the existence of mathematical structures to a few that we have to discover by means of experiments. Besides, experiments can provide us only probabilities, while mathematics tries to prove the truth of its own sentences. However, even this distinction is heavily debated among the different schools about the Foundations of Mathematics, where some mathematicians (certainly a minority), like Chaitin, affirm that Mathematics is also an experimental science. In the end, the problem is if Physics and Mathematics are both founded around the dual idea of measurement (Physics) - counting (Mathematics), and if these two ideas coincide. Of course, we can argue about foundational issues, for example, if sets are more fundamental than numbers, or more abstract or more powerful or more useful. We will see that the subtle difference between measuring (an analog process) and counting (a digital process) is a cornerstone of our number cognition.

An even more important result in Mathematical Logic could be applied here, the Tarski's theorem [4], which says: *No sufficiently powerful language is strongly-semantically-self-representational*. In other words, the problem could be that Mathematics, or more concretely, its ZFC formalization, is not powerful enough to define the semantics of Physics. Thus, in this sense, Physics would need a stronger theory than ZFC. This would be surprising, because if space and time are discrete, Finite Set Theory should suffice. But, this is surely a contentious interpretation of the Tarski's theorem.

Perhaps, a little Cognitive Neuroscience is in order to clarify the issue [13, 14], although this could not be relevant for the Platonist who beliefs that mathematical objects have an independent ontological existence. Or, perhaps, the experimental results about number cognition makes the problem even more confusing, because they point to the famous Kronecker's phrase: "God made the integers; all else is the work of man". Even it is possible that a most extreme form of intuitionistic ultrafinitism is not completely unreasonable: "God made natural numbers less than five; all else is the work of humans". In fact, numbers from 0 to 4 are powerful enough for a lot of things, for example:

- Elementary arithmetic and induction.
- The notion of even and odd numbers.
- Elementary ratios, both exact (2:1, 4:2, 3:1) and inexact (2:3, 3:2, 3:4, 4:3).
- For several reasons, the duodecimal system (base 12), which is precisely 4×3 , is the optimal number system.
- The notion of remainder.
- The notion of modular arithmetic, the important thing is not the number of cycles, but the remainder.
- A primitive or vague idea of primality. Four is divisible by two, but three is not.
- Collections with more than 4 objects can be grouped using a combination of 1 and 2 or 2 and 3 objects. Any number of divisions can be performed iteratively, therefore it is not needed to count the exact number of collections to perform a just distribution. Only the remainder is problematic.

Back to the dichotomy measuring - counting. There are two cognitive systems responsible for the representation of numbers [13], which explain the importance of the number 4:

1. *Parallel individuation system (PIS) or object tracking system (OTS)*: A non symbolic cognitive system able to represent numbers from zero to four in exact form. This system is responsible for subitizing and counting (up to four).
2. *Approximate number system (ANS)*: A cognitive system able to estimate the magnitude of a group or collection of objects without the aid of language. Accuracy improves with age in humans peaking at 15%. Curiously, infants are able to see the differences between groups with certain ratios, as 2:1, earlier than others, like 3:2, which seems to follow a logical sequence. It is an analog system which uses a compressed (log) scale whose accuracy decreases with the number of objects. Thus, we can consider this system the responsible for measuring.

Therefore, our OTS represents arithmetic (mod 4). Perhaps surprisingly arithmetic (mod 4) is key to study the properties of prime numbers, to the point that divide them in two neat sets. But the important point from our physical (dimensional) perspective is that in modular arithmetic the number of cycles is irrelevant, only the remainder matters. As we will see, this will be crucial for the extension of Physics from a linear framework to a cyclic one.

It is suggested that the ANS is heavily involved in the development of higher arithmetical (therefore, mathematical) abilities, but some problems remain to prove it conclusively. It is important to mention that the ANS shares many characteristics with other senses involved in perception, like the Weber - Fechner law and a Gaussian distribution of estimations. Thus, we can speak of a genuine "number sense" [14].

The need for larger numbers arises when you need to keep track of every object, for example, in a census. It is not surprising that the development of Mathematics coincided with the invention of writing during the first urban civilizations. While oral tradition has been robust enough to preserve as vast and complex works as the Vedas or Aboriginal Australian stories composed millennia ago, advanced Mathematics is impossible without writing, at least Mathematics as we, humans, know it. If other "Mathematics" are possible is a contentious issue. Even Turing machines and human memory are described as writing

devices. I think that this point has not been considered deeply enough in most works about the Philosophy and Foundations of Mathematics, even in its obvious educational aspects. Mathematics is so hard to learn, not because it is inherently more difficult than other disciplines (its discourse is the most logical and its results follow a definite sequence of deductive reasoning), but because it is mainly a written language, it cannot be truly read. Sure, we can simulate that we read Mathematics due to familiarity and intense training or we can translate the meaning of its symbols to higher order concepts that resemble sensory intuitions, like geometrical analogies, but the final correctness of the formalized proof is what really counts. This is the reason why there are automated theorem provers but not automated literature creators. Perhaps, some proponents of the possibility of hard Artificial Intelligence opine otherwise, but even mathematical and physical creativity is not easily explainable. One thing is to formalize a known theorem and a very different thing is to discover it and find a meaningful proof. The question is if spoken and written language are truly isomorphic. I doubt so, if only because a lot of important information, perhaps the most important, is lost in writing, like the emotions coded in intonation. Paradoxically, one of the main motivations of doing Mathematics is the deep sense of pure beauty that produces, which many think that can be explained by our evolutionary preference for symmetry.

Why is all of this relevant for Dimensional Analysis or the connection between Physics and Mathematics? The problem with Physics is that it uses at the same time concepts from spoken language and the formal machinery of Mathematics, which only makes sense in writing. Therefore, physicists have to oscillate continuously between two different worlds, causing them considerable conceptual headache. Or, from other perspective, physicists must translate sentences about measurements (analog world described by the ANS) into arithmetical statements (individual numbers described by the logical = symbolic = written extension of the OTS). This explains, from a cognitive point of view, why the mathematical formulation of Quantum Mechanics is so successful, but nobody is satisfied with its many interpretations ("shut up and calculate", really?). Thus, the Neuroscience of numerosity is important, because Physics is related with ANS while Mathematics is a logical (symbolic) extension of the OTS. Physics deals with measurements, the ratios between quantities, while Mathematics with individual structures, the quantities themselves. Interestingly, the inherent probabilistic nature of Quantum Mechanics is reminiscent of the approximate nature of the ANS estimations of the magnitude of physical collections of objects. Real measurements are

made comparing the measured object with a standard, which cannot be made with absolute precision. The role of the observer in the measurement process is evident in analog devices, where a final decision must be done about the final rounded value. However, in our digital environment, this crucial fact seems unfortunately to be sidelined or completely ignored.

Therefore, Physics can properly speak only about estimated ratios of quantities, while Mathematics describes quantities themselves. In this context, physical dimensions arise to make sense of measured ratios: *what is measured?* In this sense, dimensions are the only "exact" concepts in Physics. Length is always length and cannot be confused with mass, for example. This implies that their sum is not possible (in Linear Physics). But, we can always define new ratios, so that we can multiply or divide length and mass. If we consider dimensions *per se* exact, while their measurements not, we can identify them with natural numbers, justifying the use of coding to describe them in mathematical terms.

III. MODULAR DIMENSIONAL ANALYSIS (ARITHMETIC)

I will use modular arithmetic continuously in this work, so that it is time to explain it briefly. Modular arithmetic or clock arithmetic, because one of the clearest examples is its use in the common 12 hour division of analog clocks, studies congruence relations between natural or rational numbers. A congruence relation is an equivalence relation which allows to compare numbers using a divisibility criterion. Two numbers are congruent:

$$a \equiv b \pmod{N} \tag{1}$$

if

$$a = kN + b \tag{2}$$

so that a, b have the same remainder when they are divided by N .

The properties of physical dimensions and prime numbers are very similar in an abstract sense. Both form a kind of basis and can be seen as "orthogonal" without being formally geometrical. Thus, the idea of a Gödelian coding of physical dimensions goes beyond a mere analogy or an exercise in arithmetical representation.

If we choose a value less than 11, we have to slightly modify the numbering scheme. For example: $(L2, T3, M7, \Theta11, Q13) \pmod{5}$. At first sight there is no clear criterion to select

the value of the module, except that it must be a prime number also in order to allow the modular arithmetic of rational numbers. This is necessary because the dimensional definition of most physical variables produces rational expressions. For our present investigation only a few variables will be necessary:

- Velocity: $[v] = LT^{-1}$
- Momentum: $[p] = MLT^{-1}$
- Pressure: $[P] = ML^{-1}T^{-2}$
- Force: $[F] = MLT^{-2}$
- Energy: $[En] = ML^2T^{-2}$
- Frequency: $[f] = T^{-1}$
- Action (angular momentum): $[A] = ML^2T^{-1}$
- Electric field: $[E] = LT^{-2}MQ^{-1}$
- Magnetic flux density: $[B] = T^{-1}MQ^{-1}$
- Magnetic field strength: $[H] = L^{-1}T^{-1}Q$
- Electric displacement field: $[D] = L^{-2}Q$
- Entropy: $[S] = ML^2T^{-2}\Theta^{-1}$

In the basis $(L2, T3, M5, \Theta7, Q11) \pmod{13}$, for example, omitting $\pmod{13}$ for each term, we have, after using modular arithmetic:

- Velocity: $[v] = LT^{-1} = 2/3 \equiv 5 = M$
- Momentum: $[p] = MLT^{-1} = 10/3 \equiv 12 = L^2T$
- Pressure: $[P] = ML^{-1}T^{-2} = 5/18 \equiv 1$ (nondimensional)
- Force: $[F] = MLT^{-2} = 10/9 \equiv 4 = L^2$
- Energy: $[En] = ML^2T^{-2} = 20/9 \equiv 8 = L^3$

- Frequency: $[f] = T^{-1} = 1/3 \equiv 9 = T^2$
- Action (angular momentum): $[A] = ML^2T^{-1} = 20/3 \equiv 11 = Q$
- Electric field: $[E] = LT^{-2}MQ^{-1} = 10/99 \equiv 11 = Q$
- Magnetic flux density: $[B] = T^{-1}MQ^{-1} = 5/33 \equiv 10 = LM$
- Magnetic field strength: $[H] = L^{-1}T^{-1}Q = 11/6 \equiv 4 = L^2$
- Electric displacement field: $[D] = L^{-2}Q = 11/4 \equiv 6 = LT$
- Entropy: $[S] = ML^2T^{-2}\Theta^{-1} = 20/63 \equiv 3 = T$

This is the first basis where all dimensions can be ordered without gaps. However, in the basis $(L2, T3, M7, \Theta11, Q13) \pmod{5}$, the result is:

- Velocity: $[v] = LT^{-1} = 2/3 \equiv 4 = L^2$
- Momentum: $[p] = MLT^{-1} = 14/3 \equiv 3 = T$
- Pressure: $[P] = ML^{-1}T^{-2} = 7/18 \equiv 4 = L^2$.
- Force: $[F] = MLT^{-2} = 14/9 \equiv 1$ (nondimensional)
- Energy: $[En] = ML^2T^{-2} = 28/9 \equiv 2 = L$
- Frequency: $[f] = T^{-1} = 1/3 \equiv 2 = L$
- Action (angular momentum): $[A] = ML^2T^{-1} = 28/3 \equiv 1$ (nondimensional)
- Electric field: $[E] = LT^{-2}MQ^{-1} = 14/117 \equiv 2 = L$
- Magnetic flux density: $[B] = T^{-1}MQ^{-1} = 7/39 \equiv 3 = T$
- Magnetic field strength: $[H] = L^{-1}T^{-1}Q = 13/6 \equiv 3 = T$
- Electric displacement field: $[D] = L^{-2}Q = 13/4 \equiv 2 = L$
- Entropy: $[S] = ML^2T^{-2}\Theta^{-1} = 28/99 \equiv 2 = L$

We can easily see that the equations of Physics are dimensionally correct also in MDA. In the basis $(L2, T3, M7, \Theta11, Q13) \pmod{5}$, we have:

Maxwell's equations:

$$E/L = B/T \Rightarrow 1 = 1 \quad (3)$$

$$H/L = D/T \Rightarrow L^2 = L^2 \quad (4)$$

Uncertainty Principle:

$$pL = LT = 1 = A \quad (5)$$

$$EnT = LT = 1 = A \quad (6)$$

While in the basis $(L2, T3, M5, \Theta7, Q11) \pmod{13}$, the result is:

Maxwell's equations:

$$E/L = B/T \Rightarrow Q/L = LM/T \Rightarrow -1 = -1 \quad (7)$$

$$H/L = D/T \Rightarrow L^2/L = LT/T \Rightarrow L = L \quad (8)$$

Uncertainty Principle:

$$pL = L^2TL = L^3T = Q = A \quad (9)$$

$$EnT = L^3T = Q = A \quad (10)$$

To this point, everything seems correct. The main difference between Modular and Linear Physics is the degree of "dimensional entanglement". The cyclic nature of Modular Physics entangles all dimensions creating very interesting relations among dimensions. The crucial point is that far from being senseless, the new dimensional congruences point to very interesting coincidences with certain interpretations of Physics. For example, the basis $(L2, T3, M7, \Theta11, Q13) \pmod{5}$ resembles the geometrized units used in General Relativity, but is too poor to allow any new interpretation. However, the basis $(L2, T3, M5, \Theta7, Q11) \pmod{13}$ is rich enough to produce new results.

- Velocity and mass coincide, unifying the idea of inertia.
- The electric field is the same as charge.
- The magnetic field strength is a force, which is an area.
- The magnetic flux density is directly connected with velocity and mass. Therefore, inertia is related with magnetism. This connection is far deeper if we write the Einstein equations in dimensional form: $1/L^2 = (G/c^4)En/L^3 \Rightarrow LM = LM \Rightarrow B = B$. Therefore, B is measured as a curvature in this basis.
- Energy is the same as (3D) volume, a purely geometric quantity. We can verify that DA is consistent in this basis: $[PV] = [S\Theta] = [En]$.
- Angular momentum is equivalent to charge. Therefore, we could connect the origin of charge with spin. This coincides with some interpretations of Quantum Mechanics which relates quantum phenomena with stochastic particle properties like the zitterbewegung [15].
- Action is directly related with the intrinsic properties of particles, like spin and charge.
- Entropy and time share the same dimension. In this basis we can say that entropy and time are the same quantity, which is a very deep result.
- Momentum and the electric displacement field are entropic.

Until now, every interpretation seems plausible, because the dimensional coding is arbitrary. This would be a kind of Gödel's incompleteness result for Physics: There are infinite interpretations of Physics, none of them complete. However, a key mathematical results will show us that, in fact, the dimensional alternatives are reduced from infinite to only three meaningful bases, at least if we restrict the number of dimensions. In these bases, the discrete and the continuum can be seen as two equivalent perspectives of the same dimensional reality. Strikingly, the famous Euler formula, which relates the most important mathematical numbers: $e^{i\pi} = -1$, can be reproduced only in these bases (using reasonably low prime numbers as modules) for a concrete rational approximation of the involved transcendental numbers.

A. Complex Modular Dimensional Analysis: How the discrete and the continuum are dimensionally entangled

I will show how MDA is able to reproduce continuum results using only rational approximations. The analogies between modular arithmetic and complex analysis are not new, because modular arithmetic allows quadratic residues of -1 . The most surprising fact is that these discrete approximations are precisely those of the transcendental numbers involved in complex analysis and this correspondence only holds in three dimensional bases for a reasonably low prime module.

$(L2, T3, M7, \Theta11, Q13) \pmod{5}$:

$$\pi \simeq \frac{22}{7} \equiv 1, e \simeq \frac{11}{4} \equiv 4 \equiv -1, i \equiv 7 \Rightarrow e^{i\pi} \equiv -1 \quad (11)$$

Notice that $i \equiv 7 \equiv 2 \pmod{5}$, but $4^2 \pmod{5} = 1$. We need $i \equiv 7$ to reproduce de Euler's identity. Dimensionally: π is dimensionless, $e = L^2 = v = P$ and $i = M$.

$(L2, T3, M5, \Theta7, Q11) \pmod{13}$:

$$\pi \simeq \frac{314}{100} \equiv 6, e \simeq \frac{271}{100} \equiv 7, i \equiv 5 \Rightarrow e^{i\pi} \equiv -1 \quad (12)$$

This is a very interesting case, because it demands a finer approximation in order to fulfill the Euler's identity. Such approximation follows the decimal representation of the transcendental numbers. Dimensionally: $\pi = LT = D$, $e = L^2 = F = H$ and $i = M = v$. The dimensional interpretation of the Euler's identity is very curious in this basis:

$$\Theta^{LTM} \equiv -1 \quad (13)$$

This involves all dimensions except charge, so that this formula is a kind of "compressed" definition of all mechanical dimensions.

$(L2, T3, M5, \Theta7, Q11) \pmod{17}$:

$$\pi \simeq \frac{22}{7} \equiv 8, e \simeq \frac{11}{4} \equiv 7, i \equiv 13 \Rightarrow e^{i\pi} \equiv -1 \quad (14)$$

This will be the most powerful basis of all, as we will see, for a number of reasons. For example, in this basis the entropy is a new dimension: $S \equiv 13 \pmod{17}$. Dimensionally:

$\pi \equiv L^3 = V$, $e \equiv \Theta$ and $i \equiv S$. Due to this, the Euler's identity has a "thermodynamic" MDA interpretation.

$$\Theta^{SV} \equiv -1 \quad (15)$$

Later, we will see that it has also an "electromagnetic" interpretation. As a strange (or heterodox) as it seems, MDA permits all kinds of "forbidden" operations, like sum and exponentiation of different dimensions. But, in spite of this, all results are consistent. And this is only the beginning.

In other bases, the Euler's identity does not hold, at least for reasonably low numbers, as far as I have been able to verify. For example, (mod 7) and (mod 11) can be discarded, because $\frac{22}{7} \equiv 0 \pmod{7}$ and $\frac{11}{4} \equiv 0 \pmod{11}$. The decimal expansion approximation does not help either. Most importantly, -1 is a quadratic nonresidue in both bases, so that, we cannot have an i analogue. Only prime modules N with the property: $N \equiv 1 \pmod{4}$ have -1 as a quadratic residue, allowing us to define a dimensional "complex unity". Therefore, the next one would be 29.

$(L2, T3, M5, \Theta7, Q11) \pmod{29}$:

$$\pi \simeq \frac{22}{7} \equiv 28 \equiv -1, e \simeq \frac{11}{4} \equiv 10, i \equiv 12 \Rightarrow e^{i\pi} \equiv 1 \quad (16)$$

Even if we use $17^2 \equiv -1 \pmod{29} \Rightarrow i = 17$, the Euler's identity does not hold. Perhaps, among the infinite number of possible combinations using different approximations there is a valid one (or several), but it is notable that 5, 13 and 17 produce the correct result in an almost direct form.

Therefore, 13 and 17 are the lowest modules that allow a complete dimensional coding without gaps. 13 identifies time and entropy, while 17 produces naturally a new independent dimension for entropy. As we can see, the "complex bases" have a heavy thermodynamic flavor, pointing to the idea that Thermodynamics creates the entanglement between dimensions. This has deep connections with Quantum Mechanics through decoherence (measurement is a thermodynamic process) and even with the potential entropic origin of the fundamental forces [16, 17] and black hole thermodynamics [18]. Remember, I am not affirming that these interpretations are the right ones, or even correct from the point of view of Linear Physics. I am proving that these interpretations have merit because they arise nat-

urally from MDA and are dimensionally consistent with every experimentally valid equation of Physics. The question is if we accept MDA as a new valid approach to formalize Physics. Actually, Linear and Modular Physics are dual representations. Linear Physics achieves consistency demanding that every equation must be dimensionless in the end, while Modular Physics is consistent because everything is dimensionally congruent, even numbers have dimensions.

IV. MODULAR PHYSICS (MOD 17)

We have seen that modular physics (mod 13) has a rich interpretation. However, the Euler's identity demanded a somewhat peculiar approximation of the transcendental numbers. This same identity holds naturally (mod 17) in a simpler rational approximation: $\pi \simeq \frac{22}{7} \equiv 8 = L^3, e \simeq \frac{11}{4} \equiv 7 = \Theta, i \equiv 13 = S \Rightarrow e^{i\pi} = -1$. The dimensional list for the basis $(L2, T3, M5, \Theta7, Q11)$ (mod 17) is:

- Velocity: $[v] = LT^{-1} = 2/3 \equiv 12 = L^2T = -M$
- Momentum: $[p] = MLT^{-1} = 10/3 \equiv 9 = T^2$
- Pressure: $[P] = ML^{-1}T^{-2} = 5/18 \equiv 5 = M$
- Force: $[F] = MLT^{-2} = 10/9 \equiv 3 = T$
- Energy: $[En] = ML^2T^{-2} = 20/9 \equiv 6 = LT$
- Frequency: $[f] = T^{-1} = 1/3 \equiv 6 = LT$
- Action (angular momentum): $[A] = ML^2T^{-1} = 20/3 \equiv 1$ (nondimensional)
- Electric field: $[E] = LT^{-2}MQ^{-1} = 10/99 \equiv 8 = L^3$
- Magnetic flux density: $[B] = T^{-1}MQ^{-1} = 5/33 \equiv 12 = L^2T = v$
- Magnetic field strength: $[H] = L^{-1}T^{-1}Q = 11/6 \equiv 16 = L^4 = -1$
- Electric displacement field: $[D] = L^{-2}Q = 11/4 \equiv 7 = \Theta$
- Entropy: $[S] = ML^2T^{-2}\Theta^{-1} = 20/63 \equiv 13 = S$ (new dimension)

The dimensional interpretation of this basis is:

- Velocity is equivalent to the expansion of an area or to a negative mass. In (mod 13) velocity and mass were similarly related with the same positive sign. The magnetic flux density is directly a velocity.
- Pressure is equivalent to mass, which points to a connection with General Relativity.
- In this basis force is measured as time. Energy is just the product of length and time.
- The Planck's relation: $E = h\nu$, is direct: $E = \nu$, because the action is dimensionless, the same as angular momentum.
- A very interesting, deep and surprising result is the following. In the basis (mod 13), the action was measured as charge, indicating a connection between spin, zitterbewegung and electric charge. In this case (mod 17), the connection is similar, but with additional connections with pure geometry: $\omega = 2\pi\nu \equiv 2L^3LT \equiv Q$. Therefore, the angular frequency and the electric charge are dimensionally equivalent (mod 17). This shows that rotation is related with charge in all complex bases, which is a strong argument for linking the origin of electric charge with spin and zitterbewegung as previously commented.
- The electric field is a 3D volume, a purely geometric quantity, the same as the magnetic field strength, an hypervolume (4D). However, the electric displacement field is measured as a temperature, a thermodynamic quantity. This indicates that permittivity can be considered a macroscopic property which alters the properties of 3D volume producing heat, which is also a form of electromagnetic radiation.
- Entropy is a new fundamental dimension which is congruent to the quadratic residue of $-1 \pmod{17}$. Therefore, $S \equiv i \pmod{17}$ and we can interpret entropy as an imaginary axis.
- The Euler's identity has a curious full thermodynamic dimensional interpretation: $\Theta^{SV} \equiv -1$, which can be translated to a electromagnetic formula also: $D^{iE} \equiv -1$. Therefore, $\Theta^{SV} \equiv D^{iE}$. This suggests a deep connection between thermodynamics and electromagnetism. Notice that "impossible" relations among variables in Linear Physics are correct and meaningful in Modular Physics.

- The gravitational constant is the product of length and mass: $[G] \equiv LM$, while the dimensional Einstein's equations are: $1/L^2 = (G/c^4)En/L^3 \Rightarrow S \equiv S \Rightarrow i \equiv i$. This indicates a direct connection between Relativity, entropy and complex geometry which is conceptually consistent with the results of Linear Physics about black hole thermodynamics. Entropy is a measure of the curvature of space (mod 17) or a negative area: $S \equiv L^{-2} \equiv -L^2$. However, in Linear black hole thermodynamics entropy is proportional to the area: $S = L^2/4$ in geometrized units. In Linear Physics the sign of dimensional quantities is irrelevant, dimensions has no sign. However, in Modular Physics the sign is important and cannot be discarded. We can translate every result from Linear Physics to Modular Physics directly without problems, but we cannot do the same in reverse. The modular results are not trivially equivalent to the linear ones. Thus, the dimensional coincidence between linear black hole thermodynamics and modular entropy (mod 17) is encouraging, but the interpretation of sign is a "complex" matter.

In Modular Physics all dimensions have not the same status. Some of them are generators, producing all the other dimensions when raised to powers, with a period = mod - 1, while other dimensions have shorter periods and do not generate the rest. In this case:

- Length is not a generator: $L, L^2, L^3, L^4 \equiv -1, L^5 \equiv TM \equiv -L, L^6 \equiv S \equiv -L^2, L^7 \equiv T^2 \equiv -L^3, L^8 \equiv 1, L^9 \equiv L, \dots$
- Time is a generator: $T, T^2, T^3 \equiv LM \equiv -\Theta, T^4 \equiv S, T^5 \equiv M, T^6 \equiv TM, T^7 \equiv Q, T^8 \equiv L^4, T^9 \equiv L\Theta, T^{10} \equiv L^3, T^{11} \equiv \Theta, T^{12} \equiv L^2, T^{13} \equiv L^2T \equiv -M, T^{14} \equiv L, T^{15} \equiv LT, T^{16} \equiv 1$.
- Mass is a generator: $M, M^2 \equiv L^3, M^3 \equiv LT, M^4 \equiv S, M^5 \equiv L\Theta, M^6 \equiv L, M^7 \equiv LM, M^8 \equiv L^4, M^9 \equiv L^2T, M^{10} \equiv T^2, M^{11} \equiv Q, M^{12} \equiv L^2, M^{13} \equiv T, M^{14} \equiv TM, M^{15} \equiv \Theta, M^{16} \equiv 1$.
- Temperature is a generator: $\Theta, \Theta^2 \equiv TM, \Theta^3 \equiv T, \Theta^4 \equiv L^2, \Theta^5 \equiv Q, \Theta^6 \equiv T^2, \Theta^7 \equiv L^2T, \Theta^8 \equiv L^4, \Theta^9 \equiv LM, \Theta^{10} \equiv L, \Theta^{11} \equiv L\Theta, \Theta^{12} \equiv S, \Theta^{13} \equiv LT, \Theta^{14} \equiv L^3, \Theta^{15} \equiv M, \Theta^{16} \equiv 1$.

- Charge is a generator: $Q, Q^2 \equiv L, Q^3 \equiv M, Q^4 \equiv L^2, Q^5 \equiv LM, Q^6 \equiv L^3, Q^7 \equiv T, Q^8 \equiv L^4, Q^9 \equiv LT, Q^{10} \equiv TM, Q^{11} \equiv L^2T, Q^{12} \equiv S, Q^{13} \equiv \Theta, Q^{14} \equiv T^2, Q^{15} \equiv L\Theta, M^{16} \equiv 1$.
- Entropy is not a generator: $S, S^2 \equiv L^4 \equiv -1, S^3 \equiv L^2, S^4 \equiv 1, S^5 \equiv S, \dots$

Length and entropy are not generators (mod 17). Entropy has the shortest period = 4 and length has a period = 8. In this basis entropy is entangled directly only with areas and hypervolumes.

A. Complex dimensions

In Linear Physics we cannot define "complex" dimensions involving sums of different dimensions, but this is not a problem in Modular Physics. Complex modular dimensions allow the definition of noncommutative structures without using matrices. We will see that this is the key to obtain the equations of Physics directly without using Rayleigh monomials, although Rayleigh monomials can be used without problem also in Modular Physics. Remember that $S \equiv i$.

- $L + iL \equiv Q, L + iT \equiv \Theta, L + iM \equiv L^4, L + i\Theta \equiv L^3, L + iQ \equiv T^2, L + iS \equiv 1$.
- $T + iL \equiv L^2T, T + iT \equiv L^3, T + iM \equiv 0, T + i\Theta \equiv T^2, T + iQ \equiv LM, T + iS \equiv L$.
- $M + iL \equiv L\Theta, M + iT \equiv LM, M + iM \equiv L, M + i\Theta \equiv Q, M + iQ \equiv L^2T, M + iS \equiv L^2$.
- $\Theta + iL \equiv L^4, \Theta + iT \equiv L^2T, \Theta + iM \equiv L^2, \Theta + i\Theta \equiv S, \Theta + iQ \equiv L\Theta, \Theta + iS \equiv LT$.
- $Q + iL \equiv T, Q + iT \equiv L^4, Q + iM \equiv L^3, Q + i\Theta \equiv 0, Q + iQ \equiv 1, Q + iS \equiv LM$.
- $S + iL \equiv M, S + iT \equiv 1, S + iM \equiv LM, S + i\Theta \equiv L, S + iQ \equiv T, S + iS \equiv L^2T$.

There are very interesting congruences, for example:

- If we iterate the construction of complex dimensions, we always obtain nondimensionality ($\equiv 1$). For example: $L + iL \equiv Q, Q + iQ \equiv 1$.

- The connection between charge and rotations is obtained again in a complex form: $L + iL \equiv Q$. Additionally, a new complex entanglement among mass, temperature and charge is discovered: $M + i\Theta \equiv Q$. This suggest again that the origin of charge can be described as a stochastic phenomenon in a zitterbewegung way.
- $T + iL \equiv L^2T \equiv v \equiv B$. Therefore, we can describe velocity (and the magnetic flux density) as a complex vector involving time and length. In a sense, complex modular dimensions describe other operations as sums, in this case, a derivative, but this is not always the case, because there is not a simple substitution rule. Observe that order is crucial, which allows us to define noncommutative structures: $L + iT \equiv \Theta$. This indicates that velocity and temperature are related by a complex permutation.
- An extremely interesting result is the following. In physics (mod 13) entropy and time share the same dimension, while in the basis (mod 17) entropy is a new dimension. But, in physics (mod 17) a stronger connection is obtained, entropy is equivalent to spacetime, not only time: $L^3 + iT \equiv S$. In this way Relativity and Thermodynamics would be directly related. Furthermore, mass can be connected with entropy in higher dimensions: $L^4 + iM \equiv S$, which suggest that the use of higher spaces can result in certain unifications in a Kaluza - Klein way [19]. For electric charge this happens when $L^6 + iT^2 \equiv Q \Rightarrow S + iT^2 \equiv Q$, involving two temporal dimensions. However, we have seen that charge can be described also as a rotation and a stochastic phenomenon. Therefore, here we have a kind of physical analogue of the Gödel's incompleteness theorem for Physics: *There are many congruent physical descriptions of the same phenomenon and none of them is complete.* Each description is only a view of the complete picture and we need all of them to understand Physics fully. This is the modular version of the Dimensional Synthesis studied in Linear Physics, where the arbitrariness allowed by the II theorem and the change of basis provides different alternatives.
- 3D volume (which is congruent with π) is equivalent to a complex temporal dimension: $T + iT \equiv L^3 \equiv \pi$. Length can be described as a combination of time and entropy: $T + iS \equiv L$. In addition, length can be defined also by a complex mass, therefore: $L \equiv T + iS \equiv M + iM$.

- Remember that in physics (mod 17) entropy was not exactly proportional to an area, as in black hole thermodynamics, because a negative sign arises. However, the complex dimension formed by mass and entropy is an area: $M + iS \equiv L^2$. This is modular generalization of the Second Law.
- Thermodynamic relations arise naturally. For example: $\Theta + i\Theta = S, \Theta + iS = LT = En$. In this case, a complex dimension is equivalent to a product: $\Theta + iS \equiv \Theta S \equiv En$. Notice again, the noncommutativity: $S + i\Theta \equiv L$, another thermodynamic connection for length.
- There are not equivalent complex definitions for volume and pressure, because they are not basis dimensions: $P + iL^3 \equiv \Theta, L^3 + iP \equiv M$. However, the product holds as in Linear Physics: $PL^3 \equiv En$. However, it is suggestive that the complex combination of pressure and volume is congruent with temperature.
- Magnetic flux density can be described as a complex dimension involving mass and charge: $B \equiv M + iQ$ and the electric field is its dual (complex permutation): $E \equiv Q + iM$. This is one of the many examples of the very beautiful symmetries that Modular Physics produces. As commented before, this is not the end of the story, because $B \equiv S + iS \equiv \Theta + iT$, which points to a possible entropic origin of the magnetic phenomena. The electric field is also related with thermodynamics and complex time through $E \equiv T + iT \equiv L + i\Theta$, which again shows different symmetries between both electromagnetic fields, time and entropy.
- We can combine the electromagnetic fields to produce a new complex quantity which is congruent to charge: $F = E + iB \equiv D + iH \equiv Q$. This will be the first step to obtain the Maxwell's equations in the next section and it constitutes in itself an extraordinary result. Notice that the dual quantities have a very different meaning: $B + iE \equiv L\Theta \equiv -T, H + iD \equiv M$. However, the complex conjugates reveal new symmetries: $\bar{F} = E - iB \equiv M, D - iH \equiv T$. This point to a new interpretation where mass and even time can be considered other kind of "charges". It is noteworthy that mass does not change sign, but time does: $H + iD \equiv E - iB \equiv M, D - iH \equiv T \equiv -(B + iE)$. Amazingly, the norm is a length: $F\bar{F} \equiv E^2 + B^2 \equiv QM \equiv L^2 \Rightarrow |F| \equiv L$. This could be related with a holographic interpretation of charges, even

of fundamental particles as a kind of black holes [20]. Furthermore, remember that $L + iL \equiv M + i\Theta \equiv Q$, so that $F \equiv L + iL$, perhaps points to a string description of fields. The congruence $F \equiv M + i\Theta$ can be interpreted as describing the electric field as a pressure and the magnetic flux density as a temperature. Notice, however, that although $E + iB \equiv D + iH$ they are not equal in the sense of Linear Physics, because, for example: $|D + iH|^2 \equiv D^2 + H^2 \equiv L^4 \equiv -1$, which is just the square of $|F|^2$. Another example showing that congruence is not equality is: $E + iB \equiv EB \equiv D + iH \equiv Q$, but $DH \equiv LM$.

- Time is related with electric charge in a complex way: $T \equiv Q + iL \equiv S + iQ$. But, remember that charge is described as a rotation, so that time is connected also with spatial rotations.
- Some combinations are dimensionless: $L + iS \equiv Q + iQ \equiv S + iT \equiv 1$, describing again a complex relation among charge, length, time and entropy. Others are null, therefore they are congruent with the module: $T + iM \equiv Q + i\Theta \equiv 0$.

The use of complex dimensions reveals new rich structures and entanglements among physical quantities, many of them exhibiting symmetries. The next step is the definition of complex infinitesimal dimensions, which will allow us to obtain important equations of Physics.

V. INFINITESIMAL MODULAR DIMENSIONAL ANALYSIS

In my previous work infinitesimal dimensions in Linear Physics demanded new algebraic rules which forbade mixing of natural and infinitesimal dimensions. Formally this was satisfactory and consistent, but produced an impassable wall between the natural and the infinitesimal world that required some fixing. However, somewhat surprisingly, in Modular Physics, infinitesimals can be defined simply as rational numbers with a prime denominator, so that natural and infinitesimal quantities can be freely mixed. This will produce a new degree of "incompleteness" or arbitrariness, but this arbitrariness, instead of being harmful, will be again invaluable to reveal the full structure of physical equations.

A. Infinitesimal Modular Physics (mod 17)

The first studied basis will be the most powerful one: $(L2, T3, M5, \Theta7, Q11) \pmod{17}$. I will define infinitesimal dimensions simply as a fraction with a prime denominator. However, two questions arise immediately:

1. Which prime numbers can be used to produce infinitesimal dimensions?
2. Is there a unique way to define infinitesimals?

There are potentially (countably) infinite ways to define infinitesimals, because there are infinite prime numbers. However, we will see that the equations they produce are all equivalent, so that, the arbitrariness is not as severe as expected. The second question is more subtle. There are essentially two different ways to produce infinitesimal variables:

1. To divide every basis dimension before multiplying them to produce an infinitesimal variable.
2. To divide only the final number congruent to the variable.

The results are different and their comparison will reveal new important physical information. All of this is much better explained with the concrete examples.

1. *Infinitesimal basis defined using 31 as denominator*

The following infinitesimal dimensions are obtained:

- $d1 \equiv 1/31 \equiv Q$.
- $dL \equiv 2/31 \equiv M$.
- $dT \equiv 3/31 \equiv L^4$.
- $dM \equiv 5/31 \equiv L^2$.
- $d\Theta \equiv 7/31 \equiv T^2$.
- $dQ \equiv 11/31 \equiv L$.

- $dS \equiv 13/31 \equiv \Theta$.

Even the interpretation of these results is very interesting, but I left this for the moment in order to avoid breaking the line of reasoning.

First we can define global infinitesimals (dividing the final number): $(L2, T3, M5, \Theta7, Q11)/31 \pmod{17}$

- Velocity: $[dv] = d(LT^{-1}) = (2/3)/31 \equiv S$
- Momentum: $[dp] = d(MLT^{-1}) = (10/3)/31 \equiv L\Theta$
- Pressure: $[dP] = d(ML^{-1}T^{-2}) = (5/18)/31 \equiv L^2$
- Force: $[dF] = d(MLT^{-2}) = (10/9)/31 \equiv L^4$
- Energy: $[dEn] = d(ML^2T^{-2}) = (20/9)/31 \equiv TM$
- Frequency: $[df] = d(T^{-1}) = (1/3)/31 \equiv TM$
- Action (angular momentum): $[dA] = d(ML^2T^{-1}) = (20/3)/31 \equiv Q$
- Electric field: $[dE] = d(LT^{-2}MQ^{-1}) = (10/99)/31 \equiv T$
- Magnetic flux density: $[dB] = d(T^{-1}MQ^{-1}) = (5/33)/31 \equiv S$
- Magnetic field strength: $[dH] = d(L^{-1}T^{-1}Q) = (11/6)/31 \equiv LT$
- Electric displacement field: $[dD] = d(L^{-2}Q) = (11/4)/31 \equiv T^2$
- Entropy: $[dS] = d(ML^2T^{-2}\Theta^{-1}) = (20/63)/31 \equiv \Theta$

Notice that previous interpretations and complex symmetries are even more evident in the infinitesimal domain.

We can obtain now the Maxwell equations without much further effort. The current density is:

$$j = QT^{-1}L^{-2} \equiv L^3 \equiv E \quad (17)$$

We define the following quantities:

$$d\mathbf{F} = dE + idB \equiv T + iS \equiv L \equiv dQ \quad (18)$$

$$dtdx = dT + idL \equiv L^4 + iM \equiv S \quad (19)$$

Therefore, we obtain:

$$\frac{d\mathbf{F}}{dtdx} \equiv j \quad (20)$$

If we use a more complete complex definition of current density, we obtain the Maxwell's equations without the need of an explicit derivative:

$$J = j + iQ/L^3 \equiv Q \quad (21)$$

$$d\mathbf{F} = dJ \quad (22)$$

However, a striking property of Modular Physics is that is able to produce the correct equations without "correcting" dimensional factors. In this case:

$$J = Q + iQ/T \equiv L \quad (23)$$

$$d\mathbf{F} = J \quad (24)$$

Notice that the new J is the infinitesimal of the previous definition involving densities. This is an extraordinary and very beautiful result, because it is far from trivial, involving complex infinitesimal modular dimensions and the properties of modular arithmetic with a coding of pure numbers. Besides, the resulting equation is as compact as possible, coinciding with the inhomogeneous Maxwell equations formulated in differential forms or Clifford algebras. But, the modular interpretation is far richer, because it combines two arrows of time and its thermodynamic connection:

$$dE + idB \equiv T + iS \quad (25)$$

$$dT + idL \equiv S \quad (26)$$

$$\frac{dQ}{S} \equiv J \quad (27)$$

Using this same approach, we can obtain other important equations of Physics.

$$s = p + iL \equiv 1 \Rightarrow ds \equiv Q \quad (28)$$

$$h = En + iT \equiv Q \Rightarrow dh \equiv L \quad (29)$$

Therefore, we directly obtain:

$$ds \equiv h \quad (30)$$

which is the formal equivalent to the Hamilton - Jacobi's equation. And more impressively:

$$i \frac{ds}{dtdx} \equiv h \quad (31)$$

the Schrödinger's equation arises naturally even with the complex unity.

Besides, commutators and anticommutators can be defined, using complex permutation instead of products, with a behavior similar to its linear counterparts:

$$px = p + iL \equiv 1 \quad (32)$$

$$xp = L + ip \equiv 0 \quad (33)$$

$$[p, x] = px - xp \equiv 1 \equiv A \quad (34)$$

$$\{p, x\} = px + xp \equiv 1 \equiv A \quad (35)$$

This is a semiclassical result. Using modular dimensional anticommutators, we can put the Maxwell's equations into the form:

$$d\mathbf{F} = \{dQ, dJ\} \quad (36)$$

and even obtain a formal expression equivalent to the Dirac's equation:

$$dpm = dp + iM \equiv Q \quad (37)$$

$$dtdx = T + iL \equiv S \quad (38)$$

$$\{dpm, dtdx\} \equiv (dpm + idtdx) + (dtdx + idpm) \equiv i \quad (39)$$

where we have freely mixed natural and infinitesimal dimensions without problem. Notice that this defines a double complex structure that reminds us of a spinor. An equivalent equation, which highlights the structural similarities with the Maxwell's equations is:

$$dmdp = dM + idp \equiv L^4 \quad (40)$$

$$dtdx = T + iL \equiv S \quad (41)$$

$$\frac{dmdp}{dtdx} \equiv i \quad (42)$$

Other beautiful example is the Euler's equation for fluids:

$$\rho P = \rho + iP \equiv L^2 \Rightarrow d\rho dP = d\rho + idP \equiv LM \quad (43)$$

$$vx = v + iL \equiv L^2 \Rightarrow dvdx = dv + idL \equiv LM \quad (44)$$

$$d\rho dP \equiv dvdx \quad (45)$$

Or using (anti)commutators:

$$\{x, v\} \equiv [dP, d\rho] \quad (46)$$

Again we have a semiclassical result, reminiscent to both the Euler's equation and the Mandelung interpretation of the Schrödinger equation. To show this more clearly we can perform the following transformations:

$$\frac{v}{dL} \frac{1}{dT} \equiv 1 \Rightarrow dvdx \equiv dv + i \frac{v}{dL} v \quad (47)$$

$$i \frac{dP}{dL} \equiv \rho \Rightarrow \frac{i}{\rho} \frac{dP}{dL} \equiv 1 \quad (48)$$

$$dv + \frac{v}{dL} v \equiv 1 \Rightarrow dv + \frac{v}{dL} v \equiv \frac{i}{\rho} \frac{dP}{dL} \quad (49)$$

Notice that we have been able to pass from Lagrange's to Euler's representation using only modular operations. In order to obtain the classical Euler's equation we have to multiply the right side by i . This is possible in Linear Physics because both sides are dimensionless and $i^2 \equiv -1$ is also dimensionless, so that we can fix the Rayleigh monomial up to a dimensionless constant.

$$dv + \frac{v}{dL}v \equiv -\frac{1}{\rho} \frac{dP}{dL} \quad (50)$$

These examples would be enough to show the power of Modular Physics to reproduce the results of Linear Physics without the use of nondimensionalization. However, even deeper results can be obtained that merit further research in next works. As a hint to the possibility of a Modular Quantum Field Theory, the following creation and annihilation operators can be defined:

$$a_x = L + i/dL \equiv -1 \quad (51)$$

$$a_x^\dagger = L - i/dL \equiv M \quad (52)$$

$$[En, a_x] \equiv 1 \equiv -a_x \quad (53)$$

$$[En, a_x^\dagger] \equiv M \equiv a_x^\dagger \quad (54)$$

$$[a_x^\dagger, a_x] \equiv S \equiv i \quad (55)$$

They are not exactly equivalent to the linear ones, but they are close enough to point in the right direction.

These results are surprising in the extreme, how can a discrete coding reproduce equations involving the continuum or even infinitesimals? The answer is found in a physical equivalent of the Skolem paradox in Mathematical Logic, which says that *every countable axiomatisation of set theory in first-order logic, if it is consistent, has a model that is countable* [21]. The apparent contradiction resides in the fact that there are sentences in set theory that describe uncountable sets, so that it is possible to prove the existence of the continuum using only countable means. Therefore, for example, there are countable models of the real numbers, which are uncountable. In other words, almost all real numbers are not definable in the ZFC axiomatization of set theory, they cannot be "seen" from inside the model. The question is complex and has deep philosophical implications that unfortunately we cannot explore here, but some are evident: Do undefinable numbers really exist, or equivalently, does the continuum really exist?

In our case, Modular Physics would be a countable model of Linear Physics and the paradox is as evident as in Mathematics and even more circular. We can obtain the equations of Physics using modular dimensions, which are integer numbers in the end, but we need,

in principle, real numbers to solve them. Dimensions can be described using a discrete coding, but actual measurements of dimensions demand real numbers. However, closing the circle, in practice, every experimental measurement is a rational number with some degree of randomness (uncertainty), Quantum Mechanics point to a discrete Nature and every theoretical computation in the end produces only rational approximations of real solutions.

But, as previously commented, not only the Skolem paradox is relevant for Physics. There is also a "conceptual incompleteness", reminiscent of the Gödel's incompleteness theorems for Mathematics, in Physics, because there are potentially infinite modular models of Physics with their own interpretation. One of the sources of this arbitrariness is the possibility of using two different definitions of infinitesimal. We have seen that the global definition is able to produce very important formal equivalents of the most important equations of Physics. Now we are going to compare with the results using local infinitesimals. In order to distinguish both bases, from this point we will denote global infinitesimals as $dv, dp, \dots$ and local infinitesimals as $\delta v, \delta p, \dots$. Notice that they coincide for the basis dimensions: $dL = \delta L, dT = \delta T, \dots$

- Velocity: $[\delta v] = dLdT^{-1} \equiv M/L^4 \equiv L^2T$
- Momentum: $[\delta p] = dMdLdT^{-1} \equiv L^2M/L^4 \equiv L\Theta$
- Pressure: $[\delta P] = dMdL^{-1}dT^{-2} \equiv L^2/(ML^8) \equiv Q$
- Force: $[\delta F] = dMdLdT^{-2} \equiv L^2M/L^8 \equiv T$
- Energy: $[\delta En] = dMdL^2dT^{-2} \equiv L^2M^2/L^8 \equiv TM$
- Frequency: $[\delta f] = dT^{-1} \equiv 1/L^4 \equiv L^4$
- Action (angular momentum): $[\delta A] = dMdL^2dT^{-1} \equiv L^2M^2/L^4 \equiv L$
- Electric field: $[\delta E] = dLdT^{-2}dMdQ^{-1} \equiv ML^2/(L^8L) \equiv LM$
- Magnetic flux density: $[\delta B] = dT^{-1}dMdQ^{-1} \equiv L^2/(L^4L) \equiv TM$
- Magnetic field strength: $[\delta H] = dL^{-1}dT^{-1}dQ \equiv L/(ML^4) \equiv T$
- Electric displacement field: $[\delta D] = dL^{-2}dQ \equiv L/M^2 \equiv S$

- Entropy: $[\delta S] = dM dL^2 dT^{-2} d\Theta^{-1} \equiv L^2 M^2 / (L^8 T^2) \equiv S$

We can discover interesting results comparing with the natural and global infinitesimal bases:

- Velocity, force and entropy are natural invariants in this local basis. They produce the same result in the natural and local infinitesimal bases. This cannot be seen for any quantity in the global infinitesimal basis. This implies that: $dS \equiv d\delta S$, for example. The meaning of this fact, if any, is not clear at this point. We can suspect that velocity, time ($F \equiv T$) and entropy are "local" invariants in a modular sense, but not globally conserved quantities. How can this be interpreted?
- Only momentum and energy produce the same value in both the global and local infinitesimal bases: $dp \equiv \delta p, dEn \equiv \delta En$. Curiously, they are the only conserved quantities. The most surprising fact is that angular momentum, which shares dimensions with action, is not conserved. Therefore, in some deep sense, only purely dimensional conserved quantities are preserved between both infinitesimal bases. If this is a modular version of the Noether's theorem deserve a more careful future study.
- Notice that $P \equiv M$, while $\delta P \equiv Q$, so that it seems that charges (mass and electric) can be interpreted as some kind of pressure effect.

If we iterate the global infinitesimal operation, we obtain a cycle reproducing some different variables:

$$d1 \equiv Q \quad (56)$$

$$dQ \equiv L \quad (57)$$

$$dL \equiv M \quad (58)$$

$$dM \equiv L^2 \equiv dP \quad (59)$$

$$dL \equiv LM \equiv \delta E \quad (60)$$

$$d\delta E \equiv L^3 \equiv E \quad (61)$$

$$dE \equiv T \quad (62)$$

$$dT \equiv L^4 \equiv dF \equiv H \quad (63)$$

$$dH \equiv LT \equiv En \quad (64)$$

$$dEn \equiv TM \equiv \delta En \equiv \delta B \quad (65)$$

$$d\delta B \equiv L^2T \equiv v \equiv B \quad (66)$$

$$dB \equiv S \quad (67)$$

$$dS \equiv \Theta \quad (68)$$

$$d\Theta \equiv T^2 \equiv dD \equiv p \quad (69)$$

$$dp \equiv L\Theta \equiv \delta p \quad (70)$$

$$d\delta p \equiv 1 \quad (71)$$

This cycle has precisely a length = 16. This cycle acts as a generator, because all basis dimensions are obtained. Notice that only E and B are second order invariants: $d\delta E \equiv E, d\delta B \equiv B$, which hints to another profound structure of the electromagnetic fields. Notice that the order is important, the permuted operation δdE produce a different result,

$$\delta dE \equiv \delta T \equiv dT \equiv L^4 \equiv H \quad (72)$$

$$\delta dB \equiv \delta S \equiv dS \equiv \Theta \equiv D \quad (73)$$

which shows again a beautiful symmetry, equivalent to:

$$dE \equiv \delta H \quad (74)$$

$$dB \equiv \delta D \quad (75)$$

which, in some sense, are another set of Maxwell's equations. This also implies that there is an infinitesimal duality between E and H (global - local), and the same between B and D . If we use the symmetric representation, another duality arises:

$$\delta E \equiv -D \quad (76)$$

$$dD \equiv -E \quad (77)$$

but no similar relations hold for the magnetic fields.

A similar kind of symmetry was also present in the natural dimensions by means of complex permutations:

$$Q + iM \equiv E, M + iQ \equiv B \quad (78)$$

$$L + iD \equiv E, D + iL \equiv H \quad (79)$$

$$T + iL \equiv B, L + iT \equiv D \quad (80)$$

$$E + iD \equiv dp \quad (81)$$

$$D + iE \equiv dD \quad (82)$$

$$B + iH \equiv H \quad (83)$$

$$H + iB \equiv L \quad (84)$$

We have also: $dH \equiv En, \delta B \equiv dEn$. Other relations can be obtained using commutators:

$$[E, B] \equiv dp \quad (85)$$

$$[D, H] \equiv En \quad (86)$$

$$[dE, dB] \equiv 1 \quad (87)$$

$$[\delta E, \delta B] \equiv dD \quad (88)$$

$$[dD, dH] \equiv \delta B \quad (89)$$

$$[\delta D, \delta H] \equiv -1 \quad (90)$$

$$[E + iD, B + iH] \equiv D \quad (91)$$

$$[D + iE, H + iB] \equiv 1 \quad (92)$$

The first commutator, 85, is very similar to the Lorentz force. Actually, $B + iE \equiv dp$ also suggest a connection with the linear equation.

And some more using anticommutators:

$$\{E, B\} \equiv E \quad (93)$$

$$\{D, H\} \equiv H \quad (94)$$

$$\{dE, dB\} \equiv dE \quad (95)$$

$$\{\delta E, \delta B\} \equiv \delta E \quad (96)$$

$$\{dD, dH\} \equiv dH \quad (97)$$

$$\{\delta D, \delta H\} \equiv dE \quad (98)$$

$$\{E + iD, B + iH\} \equiv B \quad (99)$$

$$\{D + iE, H + iB\} \equiv 1 \quad (100)$$

Observe that:

$$[dp, En] \equiv Q \quad (101)$$

$$\{dp, En\} \equiv E \quad (102)$$

$$dEn + idp \equiv \delta E \quad (103)$$

The following congruences are also interesting:

$$\delta E \equiv T + iQ \quad (104)$$

$$H \equiv Q + iT \quad (105)$$

$$dD \equiv L + iQ \equiv [\delta E, \delta B] \quad (106)$$

$$dE \equiv Q + iL \equiv \{dE, dB\} \quad (107)$$

We can see that the modular structure of the electromagnetic field is incredibly rich. The Maxwell's equations in the local infinitesimal basis are:

$$\delta \mathbf{F} \equiv \delta E + i\delta B \equiv 1 \quad (108)$$

$$\delta T + i\delta L \equiv dT + idL \equiv S \quad (109)$$

$$\delta j \equiv L^2 \quad (110)$$

$$\frac{\delta \mathbf{F}}{\delta t \delta x} \equiv \delta j \quad (111)$$

This is a noteworthy result, because it reproduces the behavior of the infinitesimal dimensions of Linear Physics defined in my previous work, demanding that all dimensions be infinitesimal (substitution rule). In this case, the complex dimension: $\delta J \equiv \delta j + i\delta Q/\delta L^3$ does not produce a congruence, but this one holds:

$$j + i\delta Q/\delta L^3 \equiv 1 \Rightarrow \delta \mathbf{F} \equiv j + i\delta \rho \quad (112)$$

And even a more symmetrical one is possible with a new definition of J :

$$J = j - iQ \equiv 1 \Rightarrow \delta \mathbf{F} \equiv J \quad (113)$$

A very brief connection with Relativity, which will be explored in future works:

$$L + iT \equiv \Theta \quad (114)$$

$$L - iT \equiv L\Theta \quad (115)$$

$$T + iL \equiv v \equiv \delta v \quad (116)$$

$$T - iL \equiv Q \quad (117)$$

$$|L + iT|^2 \equiv S \quad (118)$$

$$dL + idT \equiv T^2 \quad (119)$$

$$dL - idT \equiv 1 \quad (120)$$

$$dT + idL \equiv S \quad (121)$$

$$dT - idL \equiv L \quad (122)$$

$$|dL + idT|^2 \equiv T^2 \quad (123)$$

$$p + iEn \equiv L \equiv \delta A \quad (124)$$

$$En + ip \equiv L^2 \equiv dP \quad (125)$$

$$En - ip \equiv L^3 \equiv E \quad (126)$$

$$p - iEn \equiv L^4 \equiv H \quad (127)$$

$$|En + ip|^2 \equiv dEn \quad (128)$$

$$dp + idEn \equiv \delta p + i\delta En \equiv M \quad (129)$$

$$dEn + idp \equiv \delta En + i\delta p \equiv \delta E \quad (130)$$

$$dp - idEn \equiv \delta p - i\delta En \equiv En \quad (131)$$

$$dEn - idp \equiv \delta En - i\delta p \equiv T \quad (132)$$

$$|dp + idEn|^2 \equiv S \equiv i \equiv L^{-2} \quad (133)$$

Therefore,

$$|L + iT|^2 \equiv |dp + idEn|^2 \equiv S \equiv i \equiv L^{-2} \quad (134)$$

2. Infinitesimal basis defined using 29 as denominator

We will now study the effects of changing the definition of infinitesimal using another prime with different properties: $29 \equiv 1 \pmod{4}$, in contrast with the previous $31 \equiv 3 \pmod{4}$.

- $d1 \equiv 1/29 \equiv LM$.
- $dL \equiv 2/29 \equiv T$.
- $dT \equiv 3/29 \equiv S$.
- $dM \equiv 5/29 \equiv L^4$.
- $d\Theta \equiv 7/29 \equiv L$.
- $dQ \equiv 11/29 \equiv L^3 \equiv E$.
- $dS \equiv 13/29 \equiv Q$.

Global infinitesimals (dividing the final number): $(L2, T3, M5, \Theta7, Q11)/29 \pmod{17}$

- Velocity: $[dv] = d(LT^{-1}) = (2/3)/29 \equiv 1$
- Momentum: $[dp] = d(MLT^{-1}) = (10/3)/29 \equiv M$
- Pressure: $[dP] = d(ML^{-1}T^{-2}) = (5/18)/29 \equiv L^4$
- Force: $[dF] = d(MLT^{-2}) = (10/9)/29 \equiv S$
- Energy: $[dEn] = d(ML^2T^{-2}) = (20/9)/29 \equiv T^2$
- Frequency: $[df] = d(T^{-1}) = (1/3)/29 \equiv T^2$
- Action (angular momentum): $[dA] = d(ML^2T^{-1}) = (20/3)/29 \equiv LM$
- Electric field: $[dE] = d(LT^{-2}MQ^{-1}) = (10/99)/29 \equiv L^2T$
- Magnetic flux density: $[dB] = d(T^{-1}MQ^{-1}) = (5/33)/29 \equiv 1$
- Magnetic field strength: $[dH] = d(L^{-1}T^{-1}Q) = (11/6)/29 \equiv \Theta$
- Electric displacement field: $[dD] = d(L^{-2}Q) = (11/4)/29 \equiv L$
- Entropy: $[dS] = d(ML^2T^{-2}\Theta^{-1}) = (20/63)/29 \equiv Q$

Local infinitesimals:

- Velocity: $[\delta v] = dLdT^{-1} \equiv T/S \equiv L^2T$
- Momentum: $[\delta p] = dMdLdT^{-1} \equiv L^4T/S \equiv M$
- Pressure: $[\delta P] = dMdL^{-1}dT^{-2} \equiv L^4/(TS^2) \equiv LT$
- Force: $[\delta F] = dMdLdT^{-2} \equiv L^4T/S^2 \equiv T$
- Energy: $[\delta En] = dMdL^2dT^{-2} \equiv L^4T^2/S^2 \equiv T^2$
- Frequency: $[\delta f] = dT^{-1} \equiv 1/S \equiv L^2$
- Action (angular momentum): $[\delta A] = dMdL^2dT^{-1} \equiv L^4T^2/S \equiv TM$
- Electric field: $[\delta E] = dLdT^{-2}dMdQ^{-1} \equiv TL^4/(S^2L^3) \equiv Q$

- Magnetic flux density: $[\delta B] = dT^{-1}dM dQ^{-1} \equiv L^4/(SL^3) \equiv L^3$
- Magnetic field strength: $[\delta H] = dL^{-1}dT^{-1}dQ \equiv L^3/(TS) \equiv M$
- Electric displacement field: $[\delta D] = dL^{-2}dQ \equiv L^3/T^2 \equiv L^4$
- Entropy: $[\delta S] = dM dL^2 dT^{-2} d\Theta^{-1} \equiv L^4 T^2/(S^2 L) \equiv S$

Although the concrete results are different, the general structure is the same:

- Velocity, force and entropy are natural invariants in the infinitesimal local basis.
- Only momentum and energy produce the same value in both the global and local infinitesimal bases.

It can be seen immediately that in this basis:

$$dQ \equiv E \quad (135)$$

$$dE \equiv B \quad (136)$$

$$dH \equiv D \quad (137)$$

$$\delta E \equiv Q \quad (138)$$

$$\delta B \equiv E \quad (139)$$

$$\delta D \equiv H \quad (140)$$

These relations are not the same as before (with 31), but they also exhibit an extreme degree of symmetry. Maxwell's equations:

$$j = QT^{-1}L^{-2} \equiv L^3 \equiv E \quad (141)$$

$$d\mathbf{F} = dE + idB \equiv B + i1 \equiv L^3 \equiv E \quad (142)$$

$$dtdx = dT + idL \equiv S + iT \equiv 1 \quad (143)$$

Therefore, we obtain:

$$\frac{d\mathbf{F}}{dt dx} \equiv j \quad (144)$$

which is directly equivalent to:

$$d\mathbf{F} \equiv j \quad (145)$$

Notice the consistency of the results.

Using the more complete complex definition of current density:

$$J = j + iQ/L^3 \equiv Q \Rightarrow dJ \equiv E \quad (146)$$

$$d\mathbf{F} = dJ \quad (147)$$

in complete agreement with the previous infinitesimal basis.

$$\delta\mathbf{F} \equiv \delta E + i\delta B \equiv Q + iE \equiv i \quad (148)$$

$$\delta T + i\delta L \equiv 1 \quad (149)$$

$$\delta j \equiv i \quad (150)$$

$$\frac{\delta\mathbf{F}}{\delta t \delta x} \equiv \delta j \quad (151)$$

$$\delta\mathbf{F} \equiv \delta j \quad (152)$$

However, even more powerful results can be obtained playing with the complex definitions:

$$J = Q + ij \equiv i \Rightarrow \delta\mathbf{F} \equiv J \quad (153)$$

$$\mathbf{F}^* = E - iB \equiv M \Rightarrow \mathbf{F}^* \equiv \delta p \quad (154)$$

We obtain the formal modular equivalent of the Maxwell's equation and the Lorentz force in a perfectly symmetric form. Observe that the definition of J changes compared to the previous infinitesimal basis, but only order and signs are inverted, resembling a kind of covariant - contravariant transformation. One of the most interesting results of Modular Physics is that there is no need of "correcting" dimensional factors like permittivity, permeability or velocity. The complex sums and the dimensional definition of the involved

physical quantities take care of everything in order to produce a correct equation. Although the results using infinitesimals based on 31 are very similar, the Lorentz force in such case uses a permuted but not conjugated $\mathbf{F}$. This points to transformation laws between different infinitesimal bases. In fact, there is another symmetry between the transformations of $\mathbf{F}$ and J in both bases. More food for thought.

Hamilton - Jacobi's equation:

$$s = p + iL \equiv 1 \Rightarrow ds \equiv LM \quad (155)$$

$$h = En + iT \equiv Q \Rightarrow dh \equiv E \quad (156)$$

$$ids \equiv h \quad (157)$$

Notice that, in this basis, there is not a classical result, only the quantum mechanical equation: $ds \equiv LM$, but $h \equiv Q$.

Dirac's equation:

$$dmdp = dM + idp \equiv i \quad (158)$$

$$dtdx = S + iT \equiv 1 \quad (159)$$

$$\frac{dmdp}{dtdx} \equiv i \quad (160)$$

$$dmdp \equiv i \quad (161)$$

It seems that the equations of Physics are invariant with respect to changes of infinitesimal modular bases for the same modulus. The physical interpretation changes, but the equations not. However, there are important results, like the Uncertainty Principle, that are not obtained directly as complex dimensions in all infinitesimal bases. Only certain primes act as basis for this principle. Remember that $[\hbar] \equiv 1 \pmod{17}$.

$$p + iL \equiv 1 \quad (162)$$

$$[p, L] \equiv 1 \quad (163)$$

$$d1 = \frac{1}{2} \Rightarrow \frac{p}{2} \frac{L}{2} \equiv i \Rightarrow dpdx \equiv i \quad (164)$$

$$d1 = \frac{1}{89} \Rightarrow dp + idL \equiv i \quad (165)$$

$$d1 = \frac{1}{89} \Rightarrow [dp, dx] = (dp + idL) - (dL + idp) \equiv i \quad (166)$$

This is another manifestation of the "Incompleteness of Physics".

- $d1 \equiv 1/89 \equiv S \equiv i$.
- $dL \equiv 2/89 \equiv T^2 \equiv p$.
- $dT \equiv 3/89 \equiv M \equiv P$.
- $dM \equiv 5/89 \equiv L\Theta$.
- $d\Theta \equiv 7/89 \equiv LT \equiv En$.
- $dQ \equiv 11/89 \equiv \Theta \equiv D$.
- $dS \equiv 13/89 \equiv L^4 \equiv H \equiv -1$.

For example:

$$dQ + idS \equiv E + iB \equiv D + iH \equiv Q \quad (167)$$

Is this true for the other complex basis (mod 13)?

VI. INFINITESIMAL MODULAR PHYSICS (MOD 13)

Remember that $i \equiv 5$ in this basis.

- $d1 \equiv 1/29 \equiv T^2$.
- $dL \equiv 2/29 \equiv M$.
- $dT \equiv 3/29 \equiv 1$.
- $dM \equiv 5/29 \equiv LT$.
- $d\Theta \equiv 7/29 \equiv Q$.

- $dQ \equiv 11/29 \equiv L^3$.

Global infinitesimals (dividing the final number): $(L2, T3, M5, \Theta7, Q11)/29 \pmod{13}$

- Velocity: $[dv] = d(LT^{-1}) = (2/3)/29 \equiv LT$
- Momentum: $[dp] = d(MLT^{-1}) = (10/3)/29 \equiv L^2$
- Pressure: $[dP] = d(ML^{-1}T^{-2}) = (5/18)/29 \equiv T^2$
- Force: $[dF] = d(MLT^{-2}) = (10/9)/29 \equiv LM$
- Energy: $[dEn] = d(ML^2T^{-2}) = (20/9)/29 \equiv \Theta$
- Frequency: $[df] = d(T^{-1}) = (1/3)/29 \equiv T$
- Action (angular momentum): $[dA] = d(ML^2T^{-1}) = (20/3)/29 \equiv L^3$
- Electric field: $[dE] = d(LT^{-2}MQ^{-1}) = (10/99)/29 \equiv L^3$
- Magnetic flux density: $[dB] = d(T^{-1}MQ^{-1}) = (5/33)/29 \equiv L^2T$
- Magnetic field strength: $[dH] = d(L^{-1}T^{-1}Q) = (11/6)/29 \equiv LM$
- Electric displacement field: $[dD] = d(L^{-2}Q) = (11/4)/29 \equiv L$
- Entropy: $[dS] = d(ML^2T^{-2}\Theta^{-1}) = (20/63)/29 \equiv 1$

Local infinitesimals:

- Velocity: $[\delta v] = dLdT^{-1} \equiv M$
- Momentum: $[\delta p] = dMdLdT^{-1} \equiv LTM \equiv L^2$
- Pressure: $[\delta P] = dMdL^{-1}dT^{-2} \equiv LT/M \equiv T^2$
- Force: $[\delta F] = dMdLdT^{-2} \equiv LTM \equiv L^2$
- Energy: $[\delta En] = dMdL^2dT^{-2} \equiv LTM^2 \equiv \Theta$
- Frequency: $[\delta f] = dT^{-1} \equiv 1$
- Action (angular momentum): $[\delta A] = dMdL^2dT^{-1} \equiv LTM^2 \equiv \Theta$

- Electric field: $[\delta E] = dLdT^{-2}dMdQ^{-1} \equiv MLT/L^3 \equiv \Theta$
- Magnetic flux density: $[\delta B] = dT^{-1}dMdQ^{-1} \equiv LT/L^3 \equiv L^2$
- Magnetic field strength: $[\delta H] = dL^{-1}dT^{-1}dQ \equiv L^3/M \equiv L^2T$
- Electric displacement field: $[\delta D] = dL^{-2}dQ \equiv L^3/M^2 \equiv M$
- Entropy: $[\delta S] = dMdL^2dT^{-2}d\Theta^{-1} \equiv LTM^2/Q \equiv T$

We can see that the general structure of the infinitesimals (mod 17) is preserved, but not exactly:

- Velocity, force and entropy are natural invariants in the infinitesimal local basis.
- Momentum and energy produce the same value in both the global and local infinitesimal bases, but they are not the only ones, pressure is also conserved. We will see that pressure has an unexpected role in electromagnetism.

In this basis:

$$Q \equiv E \tag{168}$$

$$dB \equiv \delta H \tag{169}$$

$$d^2D \equiv \delta D \tag{170}$$

$$\tag{171}$$

These relations are not the same as the ones produced in the basis (mod 17) and are not as symmetric. Maxwell's equations:

$$j = QT^{-1}L^{-2} \equiv L \tag{172}$$

$$d\mathbf{F} = dE + idB \equiv L^3 + iL^2T \equiv T \tag{173}$$

$$dtdx = dT + idL \equiv 1 + iM \equiv 0 \tag{174}$$

In this case, we cannot use this complex infinitesimal. Curiously, the complex current with the linear definitions also vanishes:

$$J = QL^{-3} + ij \equiv 0 \quad (175)$$

Thus, we are forced to use an alternative definition.

$$J = Q + ij \equiv L^3 \Rightarrow dJ \equiv \Theta \quad (176)$$

Therefore, this basis seems "pathological" in a sense. Interestingly, the results are in agreement with the same infinitesimal basis (mod 17) if we use $i \equiv 13$ also in this basis, which reinforces the idea that (mod 17) is a very special modular basis for the formalization of Physics. But, can we obtain the equations in this basis? We can "redeem" this basis using the other electromagnetic fields and slightly transforming some quantities.

$$d\mathbf{F}(D, H) = dD + idH \equiv L + iLM \equiv 0 \quad (177)$$

which could be interpreted as the Maxwell's equations in a vacuum. The same result is obtained in the local basis:

$$\delta\mathbf{F}(D, H) = \delta D + i\delta H \equiv M + iL^2T \equiv 0 \quad (178)$$

But, again, $\delta\mathbf{F}$ does not produce any useful results at first:

$$\delta\mathbf{F} \equiv \delta E + i\delta B \equiv \Theta + iL^2 \equiv 1 \quad (179)$$

$$\delta j \equiv M \quad (180)$$

$$\delta J \equiv \Theta \quad (181)$$

Curiously, in this case, J is also a "conserved" quantity. However, if we define: $J^* = Q - ij$:

$$\delta \mathbf{F}^* \equiv \delta E - i\delta B \equiv \Theta - iL^2 \equiv 0 \quad (182)$$

$$\delta \mathbf{F} \equiv \delta E + i\delta B \equiv \Theta + iL^2 \equiv 1 \quad (183)$$

$$J^* = Q - ij \equiv 1 \quad (184)$$

$$\delta \mathbf{F}^* \equiv 0 \quad (185)$$

$$\delta \mathbf{F} \equiv J^* \quad (186)$$

the result is correct. Something similar can be done with the global basis:

$$d\mathbf{F}^* \equiv dE - idB \equiv 0 \quad (187)$$

$$d\mathbf{F} \equiv dE + idB \equiv T \quad (188)$$

$$dtdx^* = dT - idL \equiv 1 - iM \equiv L \quad (189)$$

$$d\mathbf{F}^* \equiv 0 \quad (190)$$

$$\frac{d\mathbf{F}}{dtdx^*} \equiv J \quad (191)$$

Thus, we can recover our faith in this basis allowing these kind of transformations that we observed also in the (mod 17) infinitesimal bases. However, in this case, pressure, which is conserved, appears in the formal modular formula for the Lorentz force:

$$\mathbf{F} = E + iB \equiv dP \quad (192)$$

Interestingly:

$$E + iB \equiv dP \quad (193)$$

$$E - iB \equiv 0 \quad (194)$$

$$B + iE \equiv 0 \quad (195)$$

$$B - iE \equiv \Theta \equiv dJ \equiv dEn \quad (196)$$

Hamilton - Jacobi's equation:

$$s = p + iL \equiv T^2 \Rightarrow ds \equiv T \quad (197)$$

$$h = En + iT \equiv LM \Rightarrow dh \equiv L^2T \quad (198)$$

$$ids \equiv L \quad (199)$$

This equation does not hold, but if we make some changes as before:

$$s^* = p - iL \equiv L \Rightarrow ds^* \equiv M \quad (200)$$

$$h^* = En - iT \equiv LT \Rightarrow dh^* \equiv L \quad (201)$$

$$s^* \equiv dh^* \quad (202)$$

$$ids \equiv dh^* \quad (203)$$

Dirac's equation:

$$dmdp = dM + idp \equiv 0 \quad (204)$$

$$dtdx = dT + idL \equiv 0 \quad (205)$$

$$\frac{dmdp}{dtdx} \equiv \frac{0}{0} \quad (206)$$

$$dmdp^* = dM - idp \equiv p \quad (207)$$

$$dtdx^* = dT - idL \equiv L \quad (208)$$

$$\frac{dmdp^*}{dtdx^*} \equiv \frac{p}{L} \equiv dv \quad (209)$$

No formal analogy with the quantum mechanical equation is obtained, but it is very interesting that the first attempt ends in an indeterminate number.

Now we will use the infinitesimal basis based on 31.

- $d1 \equiv 1/31 \equiv L^3$.
- $dL \equiv 2/31 \equiv T$.
- $dT \equiv 3/31 \equiv Q$.

- $dM \equiv 5/31 \equiv 1$.
- $d\Theta \equiv 7/31 \equiv L^2$.
- $dQ \equiv 11/31 \equiv LM$.

Global infinitesimals (dividing the final number): $(L2, T3, M5, \Theta7, Q11)/31 \pmod{13}$

- Velocity: $[dv] = d(LT^{-1}) = (2/3)/31 \equiv 1$
- Momentum: $[dp] = d(MLT^{-1}) = (10/3)/31 \equiv M$
- Pressure: $[dP] = d(ML^{-1}T^{-2}) = (5/18)/31 \equiv L^3$
- Force: $[dF] = d(MLT^{-2}) = (10/9)/31 \equiv LT$
- Energy: $[dEn] = d(ML^2T^{-2}) = (20/9)/31 \equiv L^2T$
- Frequency: $[df] = d(T^{-1}) = (1/3)/31 \equiv \Theta$
- Action (angular momentum): $[dA] = d(ML^2T^{-1}) = (20/3)/31 \equiv LM$
- Electric field: $[dE] = d(LT^{-2}MQ^{-1}) = (10/99)/31 \equiv LM$
- Magnetic flux density: $[dB] = d(T^{-1}MQ^{-1}) = (5/33)/31 \equiv L$
- Magnetic field strength: $[dH] = d(L^{-1}T^{-1}Q) = (11/6)/31 \equiv LT$
- Electric displacement field: $[dD] = d(L^{-2}Q) = (11/4)/31 \equiv T^2$
- Entropy: $[dS] = d(ML^2T^{-2}\Theta^{-1}) = (20/63)/31 \equiv Q$

Local infinitesimals:

- Velocity: $[\delta v] = dLdT^{-1} \equiv T/Q \equiv M$
- Momentum: $[\delta p] = dMdLdT^{-1} \equiv T/Q \equiv M$
- Pressure: $[\delta P] = dMdL^{-1}dT^{-2} \equiv 1/(TQ^2) \equiv L^2T$
- Force: $[\delta F] = dMdLdT^{-2} \equiv T/Q^2 \equiv L^2$
- Energy: $[\delta En] = dMdL^2dT^{-2} \equiv T^2/Q^2 \equiv L^2T$

- Frequency: $[\delta f] = dT^{-1} \equiv LT$
- Action (angular momentum): $[\delta A] = dM dL^2 dT^{-1} \equiv T^2/Q \equiv L$
- Electric field: $[\delta E] = dL dT^{-2} dM dQ^{-1} \equiv T/(Q^2 LM) \equiv T$
- Magnetic flux density: $[\delta B] = dT^{-1} dM dQ^{-1} \equiv 1/(QLM) \equiv Q$
- Magnetic field strength: $[\delta H] = dL^{-1} dT^{-1} dQ \equiv LM/(TQ) \equiv \Theta$
- Electric displacement field: $[\delta D] = dL^{-2} dQ \equiv LM/T^2 \equiv L^2$
- Entropy: $[\delta S] = dM dL^2 dT^{-2} d\Theta^{-1} \equiv T^2/(Q^2 L^2) \equiv T$

We can see that the general structure of the infinitesimals is preserved:

- Velocity, force and entropy are also natural invariants in the infinitesimal local basis.
- Momentum and energy are in this case the only conserved quantities, the same as both studied infinitesimal bases (mod 17).

In this basis:

$$dE \equiv B \quad (210)$$

$$dH \equiv D \quad (211)$$

$$\delta B \equiv E \quad (212)$$

$$\delta D \equiv H \quad (213)$$

This basis is much more symmetrical than the previous one. These can be considered the Maxwell's equations if interpreted in the correct way. However, we will try to obtain them in the same way as before.

$$j = QT^{-1}L^{-2} \equiv L \quad (214)$$

$$dtdx = dT + idL \equiv Q + iT \equiv 0 \quad (215)$$

$$dtdx^* = dT - idL \equiv Q - iT \equiv T^2 \quad (216)$$

Thus, we cannot use the complex infinitesimal $dt dx$.

$$J = Q + ij \equiv L^3 \Rightarrow dJ \equiv \Theta \quad (217)$$

$$d\mathbf{F}^* \equiv dE - idB \equiv 0 \quad (218)$$

$$d\mathbf{F} = dE + idB \equiv \Theta \quad (219)$$

$$d\mathbf{F}^* \equiv 0 \quad (220)$$

$$d\mathbf{F} \equiv J \quad (221)$$

In this case, we obtain the Maxwell's equations directly without the need of $dxdt$ or its complex conjugate. It also holds that $Qd\mathbf{F} \equiv dP$.

Hamilton - Jacobi's equation:

$$s = p + iL \equiv T^2 \Rightarrow ds \equiv \Theta \quad (222)$$

$$h = En + iT \equiv LM \Rightarrow dh \equiv L \quad (223)$$

$$ids \equiv T^2 \Rightarrow ids \equiv s \quad (224)$$

$$idh \equiv LM \Rightarrow idh \equiv h \quad (225)$$

We obtain two equations, not only one, but s and h are not related. However, they are connected if we use the complex conjugates:

$$s^* = p - iL \equiv L \Rightarrow ds^* \equiv T \quad (226)$$

$$h^* = En - iT \equiv LT \Rightarrow dh^* \equiv T^2 \quad (227)$$

$$ids \equiv dh^* \quad (228)$$

$$dh \equiv s^* \quad (229)$$

Dirac's equation:

$$dmdp = dM + idp \equiv 0 \quad (230)$$

$$dtdx = dT + idL \equiv 0 \quad (231)$$

$$\frac{dmdp}{dtdx} \equiv \frac{0}{0} \quad (232)$$

$$dmdp^* = dM - idp \equiv L \quad (233)$$

$$dtdx^* = dT - idL \equiv T^2 \quad (234)$$

$$\frac{dmdp^*}{dtdx^*} \equiv LT \equiv dF \quad (235)$$

This result is more similar to a classical relativistic 4-force definition than to a quantum mechanical equation. This seems to indicate that we need more dimensions to reproduce the Dirac equation in Modular Physics than the (mod 13) basis provides, but (mod 17) is enough. Actually, (mod 13) allows L^3 and T^2 while (mod 17) is able to produce L^4 and T^2 .

VII. EFFECT OF THE CHANGE OF ORDER IN THE BASIS (MOD 17)

Until now we have seen that using a "natural" order, we can connect many important results of Linear Physics with formal equivalents in Modular Physics with different interpretations. The question now is what is the effect of changing this natural order? To answer this question we can alter the natural (mod 17) basis. For example:

$(Q2, M3, \Theta5, L7, T11) \pmod{17}$:

- Velocity: $[v] = LT^{-1} = 7/11 \equiv 13 \equiv S$
- Momentum: $[p] = MLT^{-1} = 21/11 \equiv 5 \equiv \Theta$
- Pressure: $[P] = ML^{-1}T^{-2} = 3/847 \equiv 16 \equiv Q^4$
- Force: $[F] = MLT^{-2} = 21/121 \equiv 2 \equiv Q$
- Energy: $[En] = ML^2T^{-2} = 147/121 \equiv 14 \equiv QL$
- Frequency: $[f] = T^{-1} = 1/11 \equiv 14 \equiv QL$
- Action (angular momentum): $[A] = ML^2T^{-1} = 147/11 \equiv 1$ (nondimensional)
- Electric field: $[E] = LT^{-2}MQ^{-1} = 21/242 \equiv 1$ (nondimensional)
- Magnetic flux density: $[B] = T^{-1}MQ^{-1} = 3/22 \equiv 4 \equiv Q^2$
- Magnetic field strength: $[H] = L^{-1}T^{-1}Q = 2/77 \equiv 4 \equiv Q^2$

- Electric displacement field: $[D] = L^{-2}Q = 2/49 \equiv 16 \equiv Q^4$
- Entropy: $[S] = ML^2T^{-2}\Theta^{-1} = 147/605 \equiv 13 \equiv S$ (new dimension)
- $d1 \equiv 1/29 \equiv Q\Theta$.
- $dQ \equiv 2/29 \equiv M$.
- $dM \equiv 3/29 \equiv S$.
- $d\Theta \equiv 5/29 \equiv Q^4$.
- $dL \equiv 7/29 \equiv Q$.
- $dT \equiv 11/29 \equiv Q^3$.
- $dS \equiv 13/29 \equiv T$.

Global infinitesimals (dividing the final number): $(Q2, M3, \Theta5, L7, T11)/29 \pmod{17}$

- Velocity: $[dv] = d(LT^{-1}) = (7/11)/29 \equiv T$
- Momentum: $[dp] = d(MLT^{-1}) = (21/11)/29 \equiv Q^4$
- Pressure: $[dP] = d(ML^{-1}T^{-2}) = (3/847)/29 \equiv L$
- Force: $[dF] = d(MLT^{-2}) = (21/121)/29 \equiv M$
- Energy: $[dEn] = d(ML^2T^{-2}) = (147/121)/29 \equiv Q^2$
- Frequency: $[df] = d(T^{-1}) = (1/11)/29 \equiv Q^2$
- Action (angular momentum): $[dA] = d(ML^2T^{-1}) = (147/11)/29 \equiv Q\Theta$
- Electric field: $[dE] = d(LT^{-2}MQ^{-1}) = (21/242)/29 \equiv Q\Theta$
- Magnetic flux density: $[dB] = d(T^{-1}MQ^{-1}) = (3/22)/29 \equiv QM$
- Magnetic field strength: $[dH] = d(L^{-1}T^{-1}Q) = (2/77)/29 \equiv QM$
- Electric displacement field: $[dD] = d(L^{-2}Q) = (2/49)/29 \equiv L$
- Entropy: $[dS] = d(ML^2T^{-2}\Theta^{-1}) = (147/605)/29 \equiv T$

Local infinitesimals:

- Velocity: $[\delta v] = dLdT^{-1} \equiv S$
- Momentum: $[\delta p] = dMdLdT^{-1} \equiv Q^4$
- Pressure: $[\delta P] = dMdL^{-1}dT^{-2} \equiv M^2$
- Force: $[\delta F] = dMdLdT^{-2} \equiv Q$
- Energy: $[\delta En] = dMdL^2dT^{-2} \equiv Q^2$
- Frequency: $[\delta f] = dT^{-1} \equiv M\Theta$
- Action (angular momentum): $[\delta A] = dMdL^2dT^{-1} \equiv M\Theta$
- Electric field: $[\delta E] = dLdT^{-2}dMdQ^{-1} \equiv Q^2M$
- Magnetic flux density: $[\delta B] = dT^{-1}dMdQ^{-1} \equiv QL$
- Magnetic field strength: $[\delta H] = dL^{-1}dT^{-1}dQ \equiv QL$
- Electric displacement field: $[\delta D] = dL^{-2}dQ \equiv \Theta$
- Entropy: $[\delta S] = dMdL^2dT^{-2}d\Theta^{-1} \equiv S$

The interpretation is very different, but the same general rules hold:

- Velocity, force and entropy are natural invariants in the infinitesimal local basis.
- Momentum and energy are the conserved quantities in both the global and local infinitesimal bases.

Action and entropy do not change when the order of the basis dimensions is altered. Are equations also invariant?

$$j = QT^{-1}L^{-2} \equiv M \quad (236)$$

$$dtdx = dT + idL \equiv Q^3 + iQ \equiv 0 \quad (237)$$

$$dtdx^* = dT - idL \equiv Q^3 - iQ \equiv Q^4 \quad (238)$$

Thus, we cannot use the complex infinitesimal $dt dx$.

$$J = Q + ij \equiv L \Rightarrow dJ \equiv Q \quad (239)$$

$$J^* = Q - ij \equiv QL \Rightarrow dJ^* \equiv Q^2 \quad (240)$$

$$d\mathbf{F}^* \equiv dE - idB \equiv 0 \quad (241)$$

$$d\mathbf{F} = dE + idB \equiv M \quad (242)$$

$$d\mathbf{F}^* \equiv 0 \quad (243)$$

$$d\mathbf{F} \equiv j \quad (244)$$

$$\frac{d\mathbf{F}}{dt dx^*} \equiv J^* \quad (245)$$

The Maxwell's equations are valid, using the complex conjugates.

Hamilton - Jacobi's equation:

$$s = p + iL \equiv T \Rightarrow ds \equiv Q^3 \quad (246)$$

$$h = En + iT \equiv Q^2 \Rightarrow dh \equiv QM \quad (247)$$

$$ids \equiv Q \quad (248)$$

$$idh \equiv Q\Theta \quad (249)$$

In this form, the equations do not hold. However, they are correct if we use the complex conjugates:

$$s^* = p - iL \equiv Q^4 \Rightarrow ds^* \equiv L \quad (250)$$

$$h^* = En - iT \equiv L \Rightarrow dh^* \equiv Q \quad (251)$$

$$ids \equiv dh^* \quad (252)$$

$$ids^* \equiv dh \quad (253)$$

Dirac's equation:

$$dmdp = dM + idp \equiv 0 \quad (254)$$

$$dtdx = dT + idL \equiv 0 \quad (255)$$

$$\frac{dmdp}{dtdx} \equiv \frac{0}{0} \quad (256)$$

$$dmdp^* = dM - idp \equiv M^2 \quad (257)$$

$$dtdx^* = dT - idL \equiv Q^4 \quad (258)$$

$$\frac{dmdp^*}{dtdx^*} \equiv Q^3 \equiv ds \equiv dT \quad (259)$$

Interestingly, this basis is very similar to the previously studied natural (mod 13) basis with 31 as infinitesimal denominator and the Dirac's equation cannot be obtained. Therefore, arbitrariness is not absolute, some bases have more predictive power than others.

However, we have been fortunate finding another basis where $[A] \equiv 1$ and $[S] \equiv S$. In general, this is not true, for example:

$(\Theta 2, M3, Q5, T7, L11) \pmod{17}$:

- Velocity: $[v] = LT^{-1} = 11/7 \equiv 4 \equiv \Theta^2$
- Momentum: $[p] = MLT^{-1} = 33/7 \equiv 12 \equiv \Theta^2 M$
- Pressure: $[P] = ML^{-1}T^{-2} = 3/539 \equiv 13 \equiv i$
- Force: $[F] = MLT^{-2} = 33/49 \equiv 9 \equiv M^2$
- Energy: $[En] = ML^2T^{-2} = 363/49 \equiv 14 \equiv \Theta T$
- Frequency: $[f] = T^{-1} = 1/7 \equiv 5 \equiv Q$
- Action (angular momentum): $[A] = ML^2T^{-1} = 363/7 \equiv 13 \equiv i$
- Electric field: $[E] = LT^{-2}MQ^{-1} = 33/245 \equiv 12 \equiv \Theta^2 M$
- Magnetic flux density: $[B] = T^{-1}MQ^{-1} = 3/35 \equiv 3 \equiv M$
- Magnetic field strength: $[H] = L^{-1}T^{-1}Q = 5/77 \equiv 10 \equiv \Theta Q$

- Electric displacement field: $[D] = L^{-2}Q = 5/121 \equiv 11 \equiv L$
- Entropy: $[S] = ML^2T^{-2}\Theta^{-1} = 363/98 \equiv 7 \equiv T$

This basis is completely different, but also conceptually very interesting, because the action is directly quantized: $[A] \equiv i$, and the entropy is equivalent to time.

- $d1 \equiv 1/29 \equiv \Theta Q$.
- $d\Theta \equiv 2/29 \equiv M$.
- $dM \equiv 3/29 \equiv i$.
- $dQ \equiv 5/29 \equiv \Theta^4$.
- $dT \equiv 7/29 \equiv \Theta$.
- $dL \equiv 11/29 \equiv \Theta^3$.

Hamilton - Jacobi's equation:

$$s = p + iL \equiv \Theta \Rightarrow ds \equiv M \quad (260)$$

$$s^* = p - iL \equiv Q \Rightarrow ds^* \equiv \Theta^4 \quad (261)$$

$$h = En + iT \equiv M \Rightarrow dh \equiv i \quad (262)$$

$$h^* = En - iT \equiv \Theta^3 \Rightarrow dh^* \equiv \Theta^2 M \quad (263)$$

$$ds \equiv h \quad (264)$$

$$ids \equiv s \quad (265)$$

$$idh \equiv ds^* \quad (266)$$

$$idh^* \equiv ds \quad (267)$$

Dirac's equation:

$$dmdp = dM + idp \equiv M^2 \quad (268)$$

$$dtdx = dT + idL \equiv \Theta^2 \quad (269)$$

$$\frac{dmdp}{dtdx} \equiv MQ \quad (270)$$

Even if the action is directly quantized, the Dirac's equation is not obtained in this basis. This result reinforces the idea that only certain powerful bases are able to produce formal equivalents to the quantum mechanical equations of Linear Physics involving more complex structures like spinors or commutators, but this important topic deserves a much more detailed future study.

VIII. EFFECT OF USING OTHER BASIS DIMENSIONS (THERMODYNAMIC MODULAR PHYSICS)

Finally, we can explore the consequences of using different basis dimensions. For example, we can use thermodynamic quantities, producing the basis:

$(\Theta 2, V 3, S 5, L 7, M 11, Q 13) \pmod{17}$:

- Velocity: $[v] = LT^{-1} \equiv VS$
- Momentum: $[p] = MLT^{-1} \equiv \Theta^2 V$
- Pressure: $[P] = ML^{-1}T^{-2} \equiv V^2$
- Force: $[F] = MLT^{-2} \equiv \Theta^4$
- Energy: $[En] = ML^2T^{-2} \equiv \Theta S$
- Frequency: $[f] = T^{-1} \equiv L$
- Action (angular momentum): $[A] = ML^2T^{-1} \equiv \Theta^4$
- Electric field: $[E] = LT^{-2}MQ^{-1} \equiv Q$
- Magnetic flux density: $[B] = T^{-1}MQ^{-1} \equiv \Theta$
- Magnetic field strength: $[H] = L^{-1}T^{-1}Q \equiv Q$

- Electric displacement field: $[D] = L^{-2}Q \equiv \Theta$
- Entropy: $[S] = ML^2T^{-2}\Theta^{-1} \equiv S \equiv T$

If volume, $V \equiv 3$, then $L \equiv 7$, it is not an a priori selection. Also, we obtain that $S = 5 \equiv T$. This is a very symmetric basis. For example, we obtain directly that $En \equiv \Theta S$ and gravitation is considerably simplified, because:

$$[G] = L^3M^{-1}T^{-2} \equiv 1 \quad (271)$$

$$[G/c^4] \equiv \Theta^4 \equiv -1 \quad (272)$$

And the electromagnetic fields have an extraordinary symmetry:

$$E \equiv H \quad (273)$$

$$B \equiv D \quad (274)$$

If we choose 29 as the infinitesimal generator, we have:

- $d1 \equiv 1/29 \equiv \Theta S$.
- $d\Theta \equiv 2/29 \equiv V$.
- $dV \equiv 3/29 \equiv Q$.
- $dS \equiv 5/29 \equiv \Theta^4$.
- $dL \equiv 7/29 \equiv \Theta$.
- $dM \equiv 11/29 \equiv \Theta^3$.
- $dQ \equiv 13/29 \equiv M$.

Maxwell's equations:

$$j = QT^{-1}L^{-2} \equiv \Theta L \quad (275)$$

$$dtdx = dT + idL \equiv \Theta^2 V \quad (276)$$

$$dtdx^* = dT - idL \equiv \Theta \quad (277)$$

$$J = Q + ij \equiv \Theta^3 \quad (278)$$

$$J^* = Q - ij \equiv 1 \quad (279)$$

$$d\mathbf{F} = dE + idB \equiv \Theta^4 \quad (280)$$

$$d\mathbf{F}^* \equiv dE - idB \equiv \Theta V \quad (281)$$

$$\frac{d\mathbf{F}}{dtdx^*} \equiv J \quad (282)$$

Using the alternative definition: $J_1 = Q + iQT^{-1} \equiv \Theta V$, we obtain:

$$d\mathbf{F}^* \equiv J_1 \quad (283)$$

Hamilton - Jacobi's equation:

$$s = p + iL \equiv 1 \Rightarrow ds \equiv \Theta S \quad (284)$$

$$s^* = p - iL \equiv \Theta V \Rightarrow ds^* \equiv V^2 \quad (285)$$

$$h = En + iT \equiv L \Rightarrow dh \equiv \Theta \quad (286)$$

$$h^* = En - iT \equiv Q \Rightarrow dh^* \equiv M \quad (287)$$

$$ids \equiv dh^* \quad (288)$$

$$idh \equiv ds^* \quad (289)$$

$$idh^* \equiv h \quad (290)$$

Dirac's equation:

$$dpdm = dp + idM \equiv V \quad (291)$$

$$dtdx = dT + idL \equiv \Theta^2 V \quad (292)$$

$$\frac{dpdm}{dtdx} \equiv i \quad (293)$$

If we choose 31 as the infinitesimal generator, we have:

- $d1 \equiv 1/31 \equiv M$.
- $d\Theta \equiv 2/31 \equiv S$.
- $dV \equiv 3/31 \equiv \Theta^4$.
- $dS \equiv 5/31 \equiv \Theta^2$.
- $dL \equiv 7/31 \equiv V^2$.
- $dM \equiv 11/31 \equiv \Theta$.
- $dQ \equiv 13/31 \equiv L$.

Maxwell's equations:

$$j = QT^{-1}L^{-2} \equiv \Theta L \quad (294)$$

$$dtdx = dT + idL \equiv V \quad (295)$$

$$dtdx^* = dT - idL \equiv V^2 \quad (296)$$

$$J = Q + ij \equiv \Theta^3 \quad (297)$$

$$J^* = Q - ij \equiv 1 \quad (298)$$

Interestingly, in this case the alternative definitions are:

$$J_1 = Q + iQT^{-1} \equiv \Theta V \quad (299)$$

$$J_2 = QL^{-3} + ij \equiv S \quad (300)$$

But only the previous definition of J produces the correct Maxwell's equations.

$$d\mathbf{F} = dE + idB \equiv \Theta^2 \quad (301)$$

$$d\mathbf{F}^* \equiv dE - idB \equiv \Theta S \quad (302)$$

$$\frac{d\mathbf{F}}{dtdx^*} \equiv J \quad (303)$$

This strange phenomenon demanding certain combination of charge and charge densities to obtain the Maxwell's equations in each basis will be studied in the future.

Hamilton - Jacobi's equation:

$$s = p + iL \equiv 1 \Rightarrow ds \equiv M \quad (304)$$

$$s^* = p - iL \equiv \Theta V \Rightarrow ds^* \equiv VS \quad (305)$$

$$h = En + iT \equiv L \Rightarrow dh \equiv V^2 \quad (306)$$

$$h^* = En - iT \equiv Q \Rightarrow dh^* \equiv L \quad (307)$$

$$ids \equiv h \quad (308)$$

$$dh^* \equiv h \quad (309)$$

$$idh \equiv ds^* \quad (310)$$

$$idh^* \equiv s^* \quad (311)$$

Dirac's equation:

$$dpdm = dp + idM \equiv S \quad (312)$$

$$dtdx = dT + idL \equiv V \quad (313)$$

$$\frac{dpdm}{dtdx} \equiv i \quad (314)$$

IX. CONCLUSION

Linear Physics is based on dimensional balance, all valid equations must be dimensionless. Therefore, many mathematical operations cannot be done with dimensions, like

sums. Besides, physical constants and conversion factors are necessary to produce the correct dimensions in equations. In the linear approach dimensions can be combined only as monomials, which can be interpreted as braids. This would be a "tensorial" formulation of Physics. However, a new kind of Physics is possible, where dimensional braids can be twisted, creating a conceptual equivalent to a "spinorial" formulation. This Modular Physics is not merely a reformulation in terms of a different mathematical structure. It goes down to the heart of Mathematics, coding dimensions as prime numbers, using the same Gödel's idea that paved the way to many of the most important theorems in Mathematical Logic. Modular arithmetic is also a fundamental tool in Number Theory. The key idea is to create equivalence classes of numbers based on their remainders when divided by some other number. This produces cyclic structures where the number of cycles is irrelevant, only the remainder is meaningful. When these ideas are applied to physical variables defined by their dimensional content, the result is a cyclic entanglement of dimensions that produce striking consequences.

We could expect that Modular Physics would be completely arbitrary. There are infinite prime numbers and the potential dimensional bases are also infinite. Besides, from a conceptual point of view, dimensions are just opposite pure numbers. Thus, how is it possible to obtain any reasonable information transforming dimensions into pure numbers? Furthermore, pure numbers can be operated without the severe restrictions imposed to dimensions. Has any sense to allow unrestricted arithmetic in Dimensional Analysis? Defying our intuition, the Gödel's coding works also in Physics and modular arithmetic produces amazingly deep results.

Modular dimensional bases are not as arbitrary as expected. Only three of them allow a simple modular equivalent to Complex Analysis using the rational approximations of the important transcendental numbers. The immediate consequence is that not only dimensions can be coded as numbers, but also numbers can be coded as dimensions. The existence of a "complex unit" permits the definition of "complex dimensions", which allow the construction of higher algebraic structures, even noncommutative ones. These structures involving the sum of different dimensions are unthinkable in Linear Physics, but in Modular Physics they are not only completely natural, but the key to obtain the equations of Physics in their most pure form, even without dimensional correcting factors. How is this possible? Due to its cyclic nature, Modular Physics has an extremely rich structure full of symmetries and

the complex structures take care of everything necessary to substitute dimensional balance (nondimensionality) by simple arithmetical congruences in a transparent way.

The equations of Physics are, in general, differential equations, needing some definition of infinitesimals. Surprisingly, we can define infinitesimals in Modular Physics simply dividing all variables or dimensions by the same prime number. Different "infinitesimals" produce different results, but by simple complex conjugations and permutations some important equations are always obtained, like the Maxwell's equations. This points to a deeper algebra that will be studied in future works. However, this is not the end of the story. In Dimensional Analysis arbitrariness = richness and this case is not an exception. We can define infinitesimal physical quantities in two different ways. The first one is a global infinitesimal obtained by dividing the final value of a variable by the prime generator of the infinitesimals. The second one is constructed substituting each basis dimension by its infinitesimal value. As expected, almost all infinitesimal quantities produce different values for their corresponding global and local representations. But the linear conserved quantities: linear momentum (but not angular momentum, whose dimensions are the same as the action, which is not conserved) and energy are the same in both cases in a striking parallel with Noether's theorems, but more constrained. Some infinitesimal modular bases seem more powerful than others, because equations involving more complex structures, like the Dirac's equation or quantum commutators are not obtained. This also deserves more future research.

Finally, Modular Physics, due to its peculiar arbitrariness (richness) allows the formulation of the problem of the conceptual interpretation of physical theories, like Quantum Mechanics, from another perspective. Many different modular bases can be used which produce the same (or formally equivalent) equations, but their dimensional congruences are very different. For example, in some bases time and entropy are equivalent, but in others entropy is a new dimension and even can be made congruent to the "complex unity" in such bases. Other examples are the possibility of dimensional "unification" of the concepts of spin, zitterbewegung, charge, electromagnetic fields and even Thermodynamics. Therefore, from the point of view of Modular Physics a stochastic interpretation of Quantum Mechanics is as valid as an information based one, for example. Of course, it would be possible that, in the end, only a few bases are able to produce all relevant equations of Physics, thus constraining also the interpretations, but the contrary is also possible, that some algebra allows the transformation from one modular basis to any other, and potentially infinite

interpretations can be considered.

Finally, is Modular Physics only a "curiosity" to produce formally similar known results in Linear Physics? Is it only a tool to formalize Physics in a new way to study it using tools from Mathematical Logic and Number Theory? Perhaps, Modular Physics is more powerful than that and will open new insights, being as useful for Physics as modular structures are for Mathematics. If so, time will say. Anyway, to obtain the Maxwell's equations from pure numbers, even if we only accept these results as "strange congruences", has been a most satisfying and beautiful experience.

Acknowledgement

I want to express my deepest gratitude to my friends and colleagues José Luis Fernández and Javier Rodríguez Laguna for his generous reviews.

-
- [1] Martínez Rojas J. A. (2023). "Foundations of Differential and Integral Dimensional Analysis: A new answer to the Wigner's question *The Unreasonable Effectiveness of Mathematics in the Natural Sciences*". Preprint version.
 - [2] Wigner E. P. (1960). "The unreasonable effectiveness of mathematics in the natural sciences", Richard Courant lecture in mathematical sciences delivered at New York University, May 11, 1959. Communications on Pure and Applied Mathematics, XIII, 001-14.
 - [3] Nagel, E., & Newman, J. R. (2012). "Godel's proof". Routledge.
 - [4] Rautenberg, W. (2006). "A concise introduction to mathematical logic". New York, NY: Springer.
 - [5] Bridges D., Palmgren E. & Ishihara H. (2022). "Constructive Mathematics". <https://plato.stanford.edu/entries/mathematics-constructive/> (Accessed February 2023).
 - [6] Simpson, S. G., & Simpson, S. G. (2009). "Subsystems of second order arithmetic (Vol. 1)". Cambridge University Press.
 - [7] Baker, A. (2012). "A comprehensive course in number theory". Cambridge University Press.
 - [8] Ireland, K., Rosen, M. I., & Rosen, M. (1990). "A classical introduction to modern number theory (Vol. 84)". Springer Science & Business Media.

- [9] Apostol, T. M. (1998). "Introduction to analytic number theory". Springer Science & Business Media.
- [10] Fürer, M. (2007, June). "Faster integer multiplication". In Proceedings of the thirty-ninth annual ACM symposium on Theory of computing (pp. 57-66).
- [11] Koblitz, N. (1994). "A course in number theory and cryptography (Vol. 114)". Springer Science & Business Media.
- [12] <https://en.wikipedia.org/wiki/Tangloids>(Accessed February 2023).
- [13] Nieder, A. (2005). "Counting on neurons: the neurobiology of numerical competence". *Nature Reviews Neuroscience*, 6(3), 177-190.
- [14] Dehaene, S. (2011). "The number sense: How the mind creates mathematics". OUP USA.
- [15] Hestenes, D. (1990). The zitterbewegung interpretation of quantum mechanics. *Foundations of Physics*, 20(10), 1213-1232.
- [16] Padmanabhan, T. (2010). Thermodynamical aspects of gravity: new insights. *Reports on Progress in Physics*, 73(4), 046901.
- [17] Verlinde, E. (2011). On the Origin of Gravity and the Laws of Newton. *Journal of High Energy Physics*, 2011(4), 1-27.
- [18] Carlip, S. (2014). Black hole thermodynamics. *International Journal of Modern Physics D*, 23(11), 1430023.
- [19] Overduin, J. M., & Wesson, P. S. (1997). Kaluza-klein gravity. *Physics reports*, 283(5-6), 303-378.
- [20] Burinskii, A. (2008). The Dirac-Kerr-Newman electron. *Gravitation and Cosmology*, 14, 109-122.
- [21] Bays, T. (2000). Reflections on Skolem's paradox. University of California, Los Angeles.